ELL 100 – Introduction to Electrical Engineering

Lecture 30: Digital Circuits

Boolean Algebra

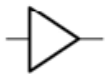
(Analysis of Binary Logic Functions)

Outline

- Rules of Combinational Logic (Laws of Boolean Algebra)
- Half & Full Adder and Subtractor circuits
- SOP (Sum of Products) and POS (Product of Sums) forms of logical expressions
- Karnaugh Maps for simplifying logical expressions

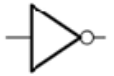
Recap of Various Logic Gates

YES



INPUT	OUTPUT
A	
0	0
1	1

NOT



INPUT	OUTPUT
A	
0	1
1	0

AND



INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

OR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

XOR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

NAND



INPUT		OUTPUT
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

NOR



INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

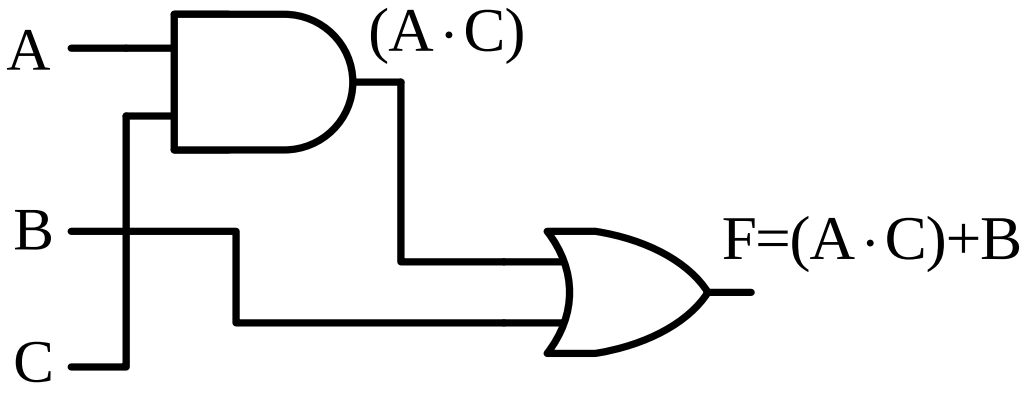
XNOR

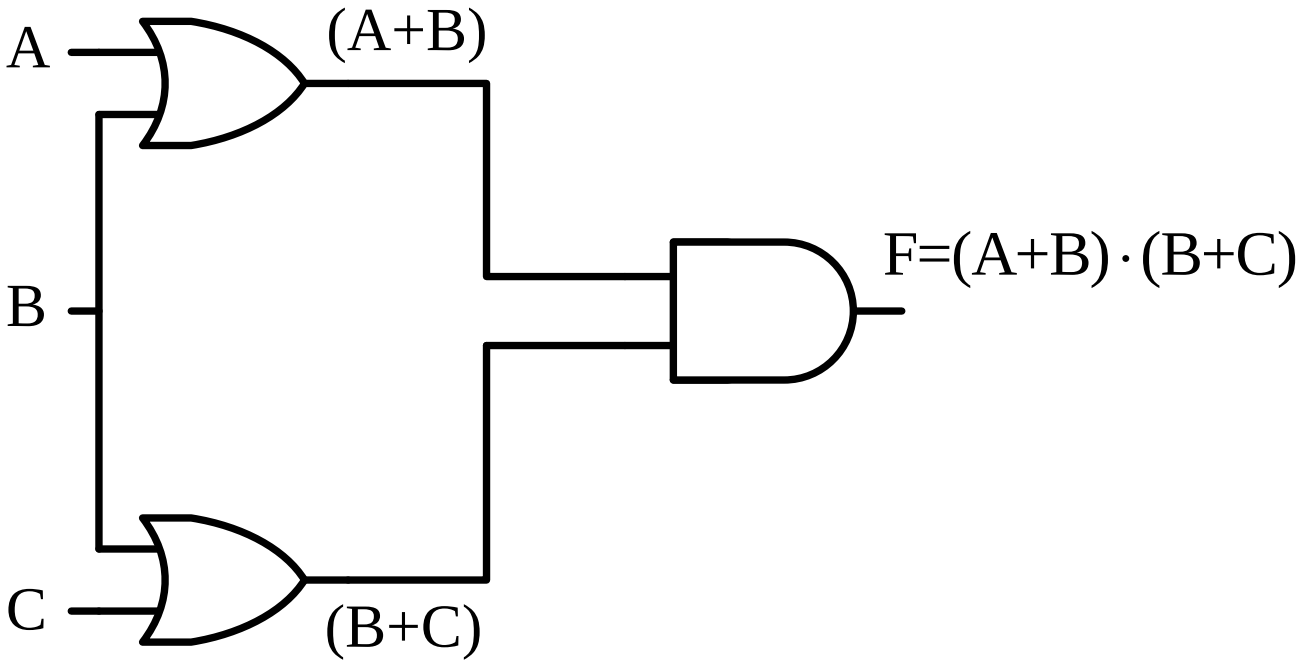


INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	1

Logic Networks

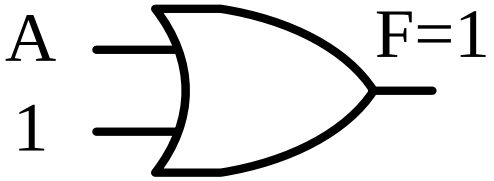
Logic gates can be interconnected to give a wide variety of functions.



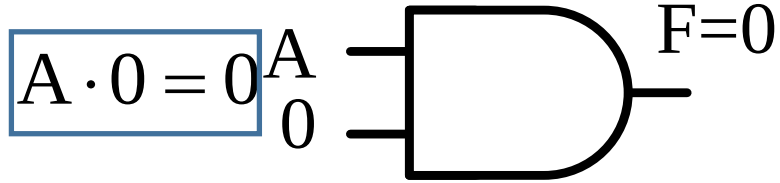


BOOLEAN IDENTITIES

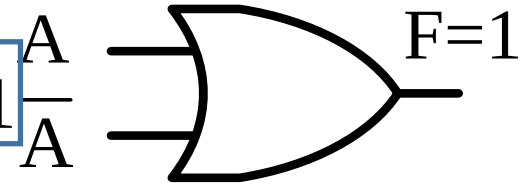
$A+1=1$



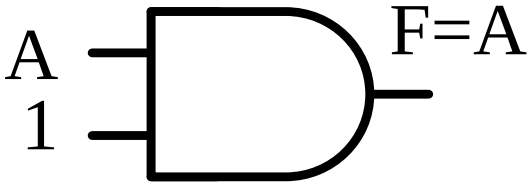
$A \cdot 0 = 0$



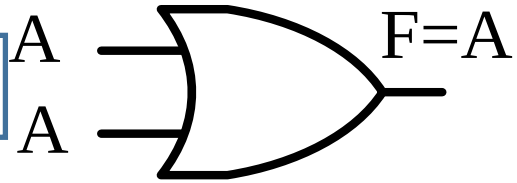
$A+\overline{A}=1$



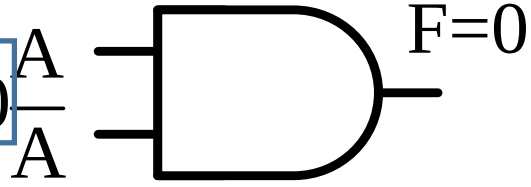
$A \cdot 1=A$



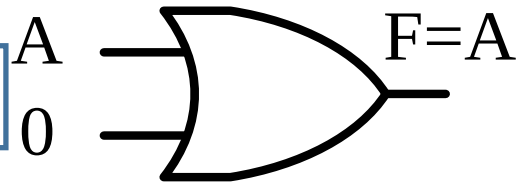
$A+A=A$



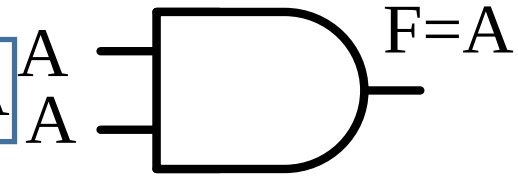
$A \cdot \overline{A}=0$



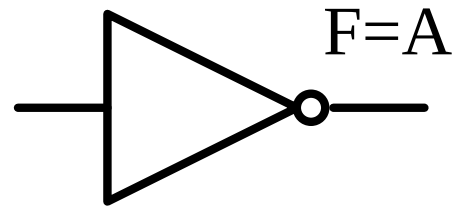
$A+0=A$



$A \cdot A=A$



$\overline{\overline{A}}=A$



RULES OF COMBINATIONAL LOGIC

Laws of Boolean Algebra ('+' denotes 'OR', '**AB**' denotes '**A** AND **B**')

a) Commutative rules: $A+B = B+A$; $AB = BA$

b) Associative rules: $A+(B+C) = (A+B)+C$; $A(BC) = (AB)C$

c) Distributive rules: $A+BC = (A+B)(A+C)$; $A(B+C) = AB+AC$

$A+AB = A$; $A(A+B) = A$

d) de Morgan's laws: $\overline{A \cdot B \cdot C} = \overline{A}+\overline{B}+\overline{C}$; $\overline{A} \cdot \overline{B} \cdot \overline{C} = \overline{A+B+C}$

RULES OF COMBINATIONAL LOGIC

Verification of distributive rule A+BC = (A+B)(A+C) via truth table

A	B	C	BC	A+BC	A+B	A+C	(A+B)(A+C)
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

Laws of Boolean Algebra used for simplifying logical expressions

($A^l = \bar{A}$ denotes the complement/inverse/NOT of A)

$$A+0=A$$

$$A+1=A$$

$$A\cdot 0=0$$

$$A\cdot 1=A$$

$$A+A=A$$

$$A+A^l=1$$

$$A\cdot A=A$$

$$A\cdot A^l=0$$

$$(A^l)^l=A$$

$$A+AB=A$$

$$A+A^lB=A+B$$

$$A+B=B+A$$

$$A\cdot B=B\cdot A$$

$$\overline{A+B}=\overline{A}\cdot\overline{B}$$

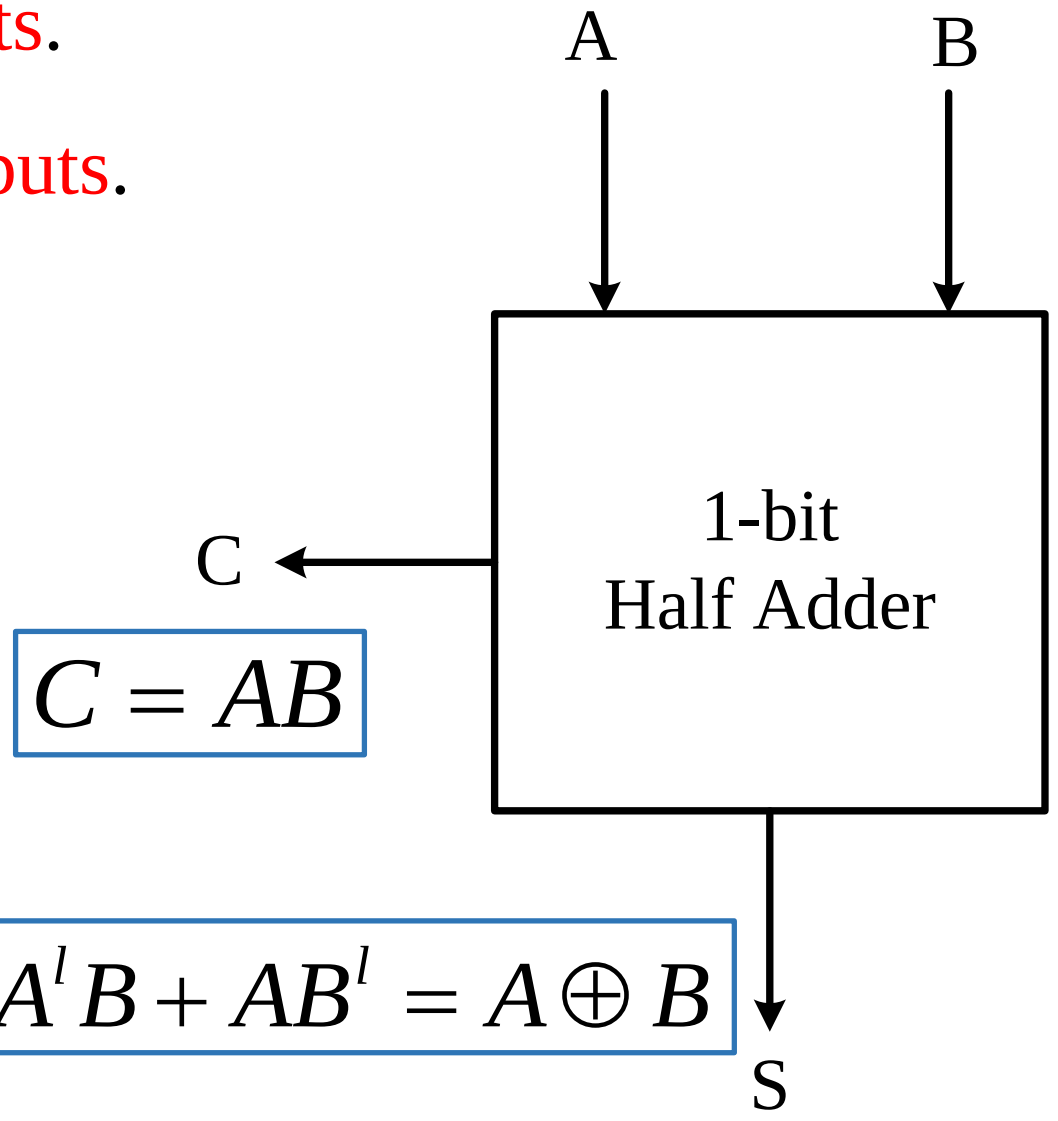
$$\overline{A\cdot B}=\overline{A}+\overline{B}$$

$$(A+B)(A+C)=A+BC$$

HALF ADDER CIRCUIT

- Takes two binary digits ***A*** and ***B*** as inputs.
- Gives out sum ***S*** and carry ***C*** bits as outputs.

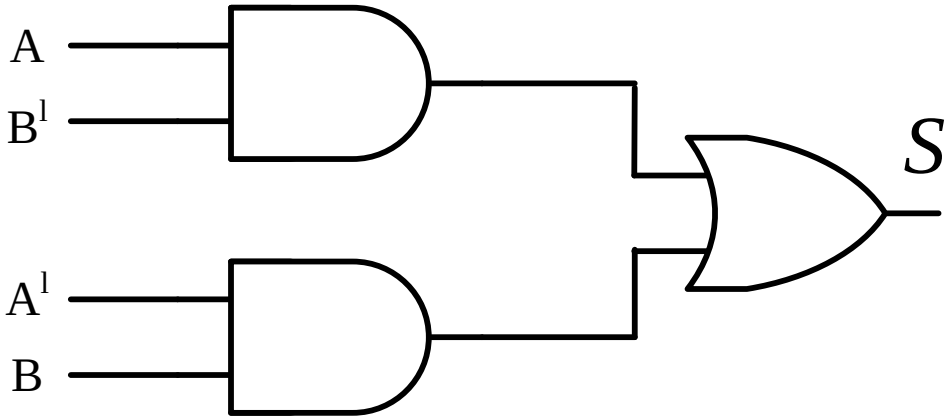
A	*B*	*C*	*S*
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



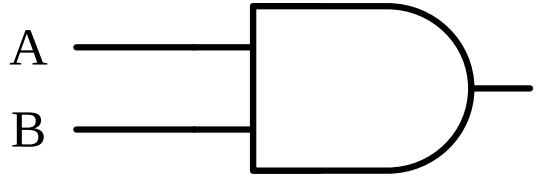
$$C = AB$$

$$S = A^{l}B + AB^{l} = A \oplus B$$

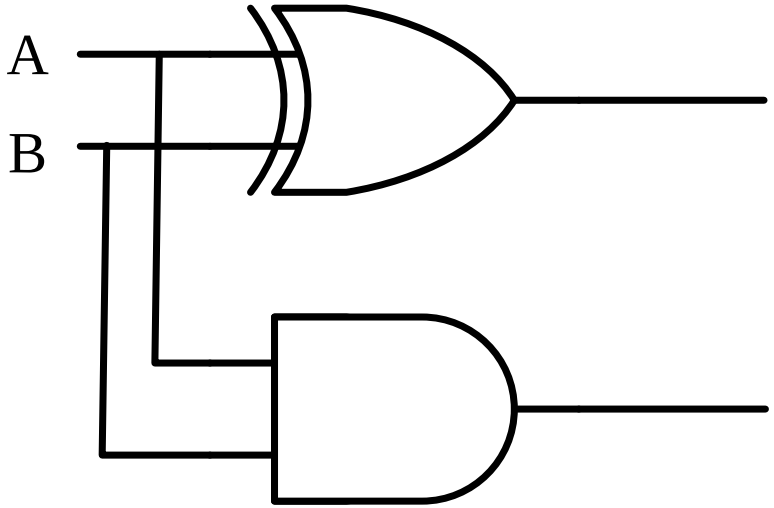
HALF ADDER CIRCUIT



$$S = A^l B + A B^l$$



$$C = AB$$



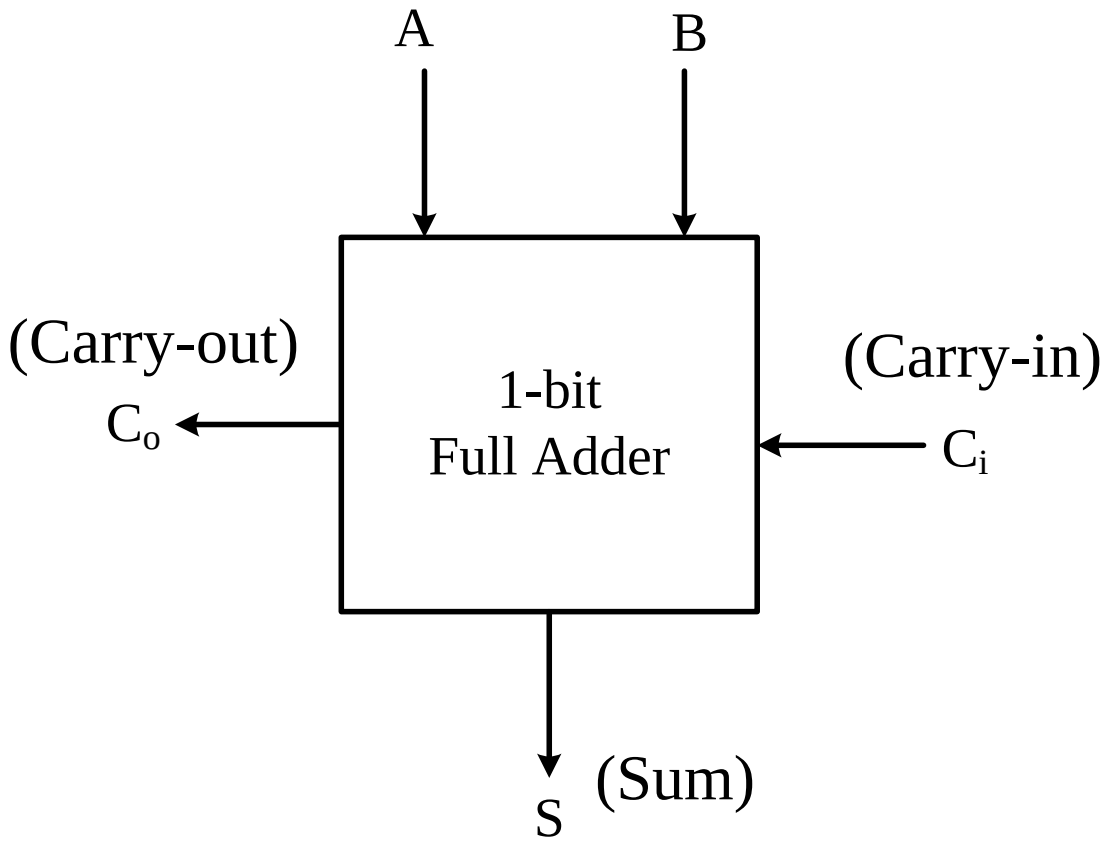
$$S = A \oplus B$$

$$C = AB$$

FULL ADDER CIRCUIT

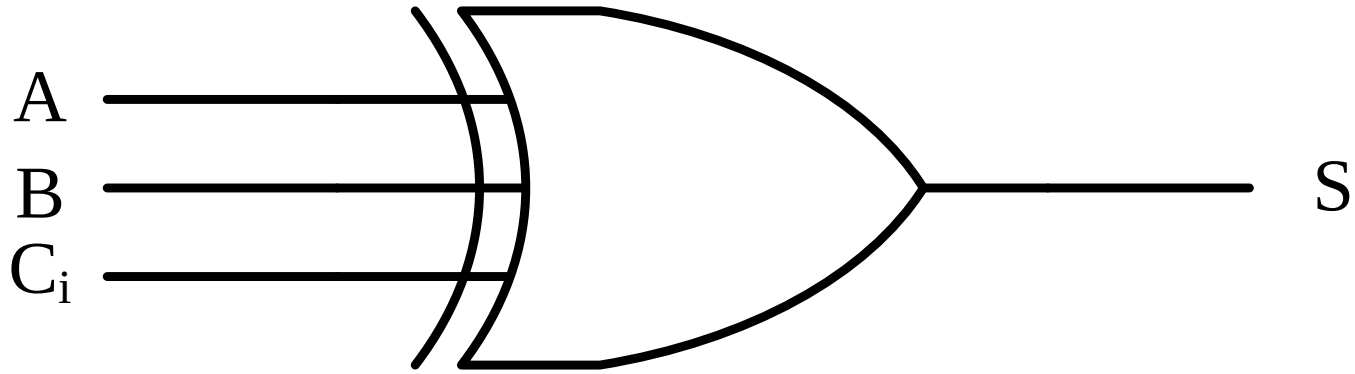
- Half Adder accepts only 2 input bits. If two n-bit binary numbers are to be added, there can be a carry bit in various places also.
- **Full Adder** makes provision for a carry-in bit also i.e. total 3 input bits.

A	B	C_i	C_o	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



FULL ADDER CIRCUIT (SUM)

A	B	C_i	C_o	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

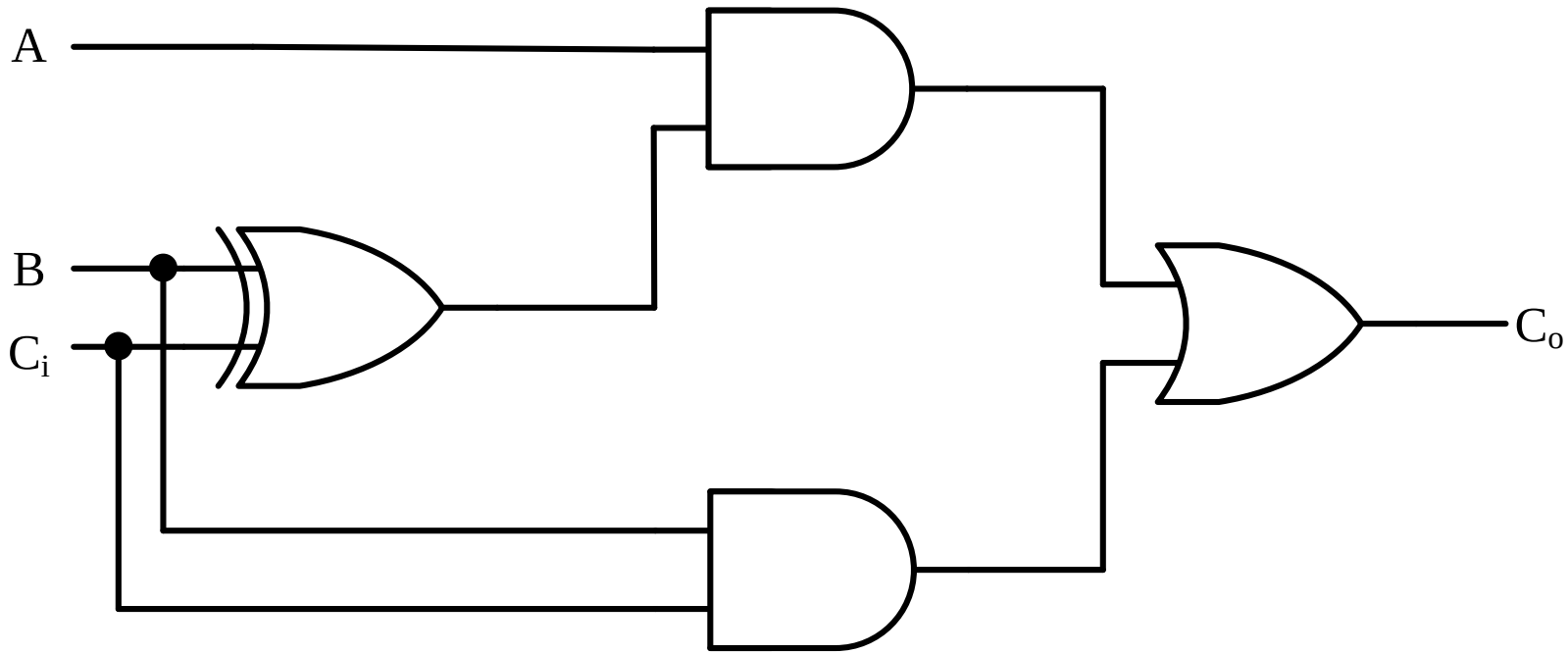


$$S = A^{l}B^{l}C_i + A^{l}BC_i^{\,l} + AB^{l}C_i^{\,l} + ABC_i = A^{l}(B^{l}C_i + BC_i^{\,l}) + A(B^{l}C_i^{\,l} + BC_i)$$

$$S = A^{l}\left(B \oplus C_i\right) + A\left(\overline{B \oplus C_i}\right) = A \oplus B \oplus C_i$$

FULL ADDER CIRCUIT (CARRY OUT)

A	B	C_i	C_o	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



$$C_o = A^l BC_i + AB^l C_i + ABC_i^l + ABC_i = BC_i(A + A^l) + A(BC_i^l + B^l C_i)$$

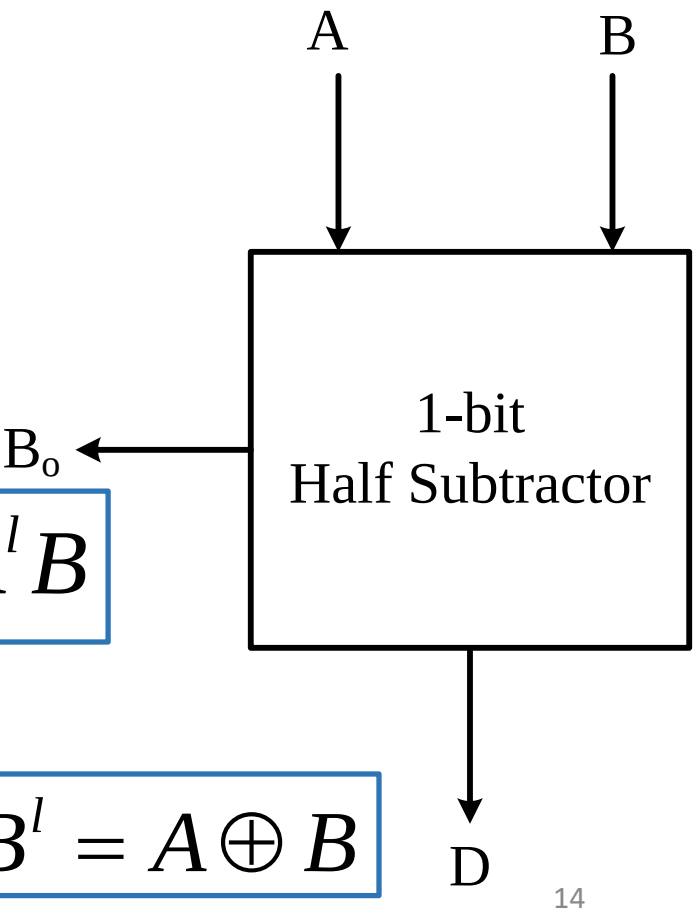
$$C_o = BC_i + A(B \oplus C_i)$$

HALF SUBTRACTOR CIRCUIT

Performs subtraction of two bits, one is minuend **A** and other is subtrahend **B**

Has two inputs and two outputs. The two output bits, **D** and $\mathbf{B_o}$ represent the **difference** and **output borrow**, respectively.

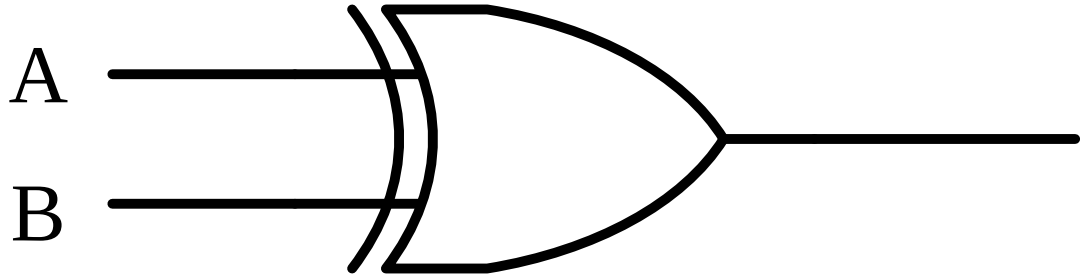
A	B	D	B_o
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



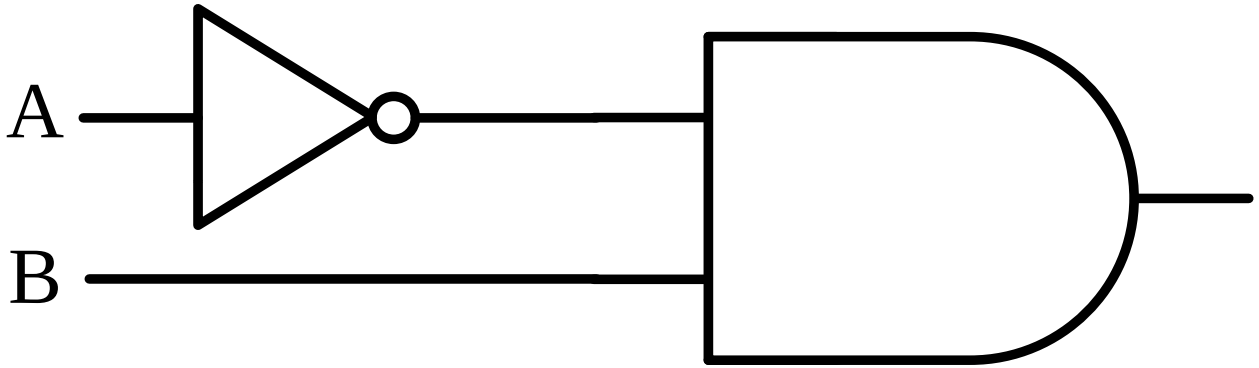
$$B_o = A^l B$$

$$D = A^l B + A B^l = A \oplus B$$

HALF SUBTRACTOR CIRCUIT



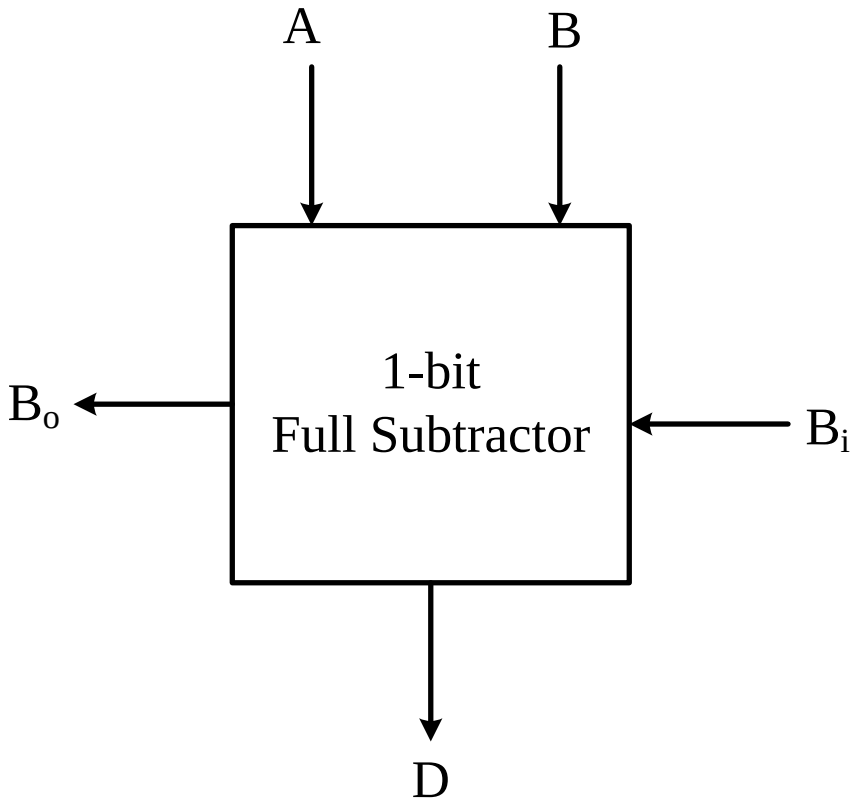
$$D = A^{l}B + AB^{l} = A \oplus B$$



$$B_o = A^{l}B$$

FULL SUBTRACTOR CIRCUIT

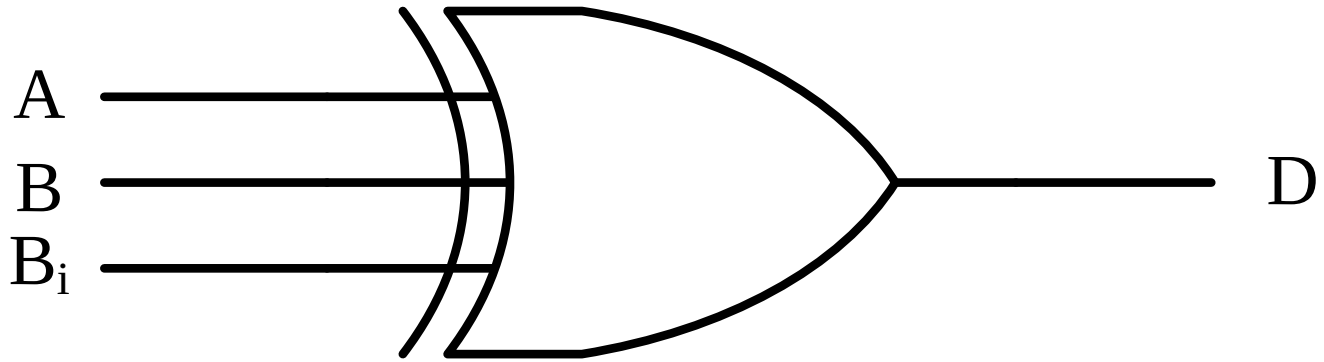
Performs subtraction of two bits, taking into account borrow of the previous adjacent lower bit also. Has three inputs **A**, **B** and $\mathbf{B_i}$, denoting the **minuend**, **subtrahend**, and **previous borrow**, respectively. The two outputs, **D** and $\mathbf{B_o}$ represent the **difference** and **output borrow**.



A	B	B_i	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

FULL SUBTRACTOR CIRCUIT (DIFFERENCE)

A	B	B_i	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



$$D = A^l B^l B_i + A^l B B_i^l + A B^l B_i^l + A B B_i$$

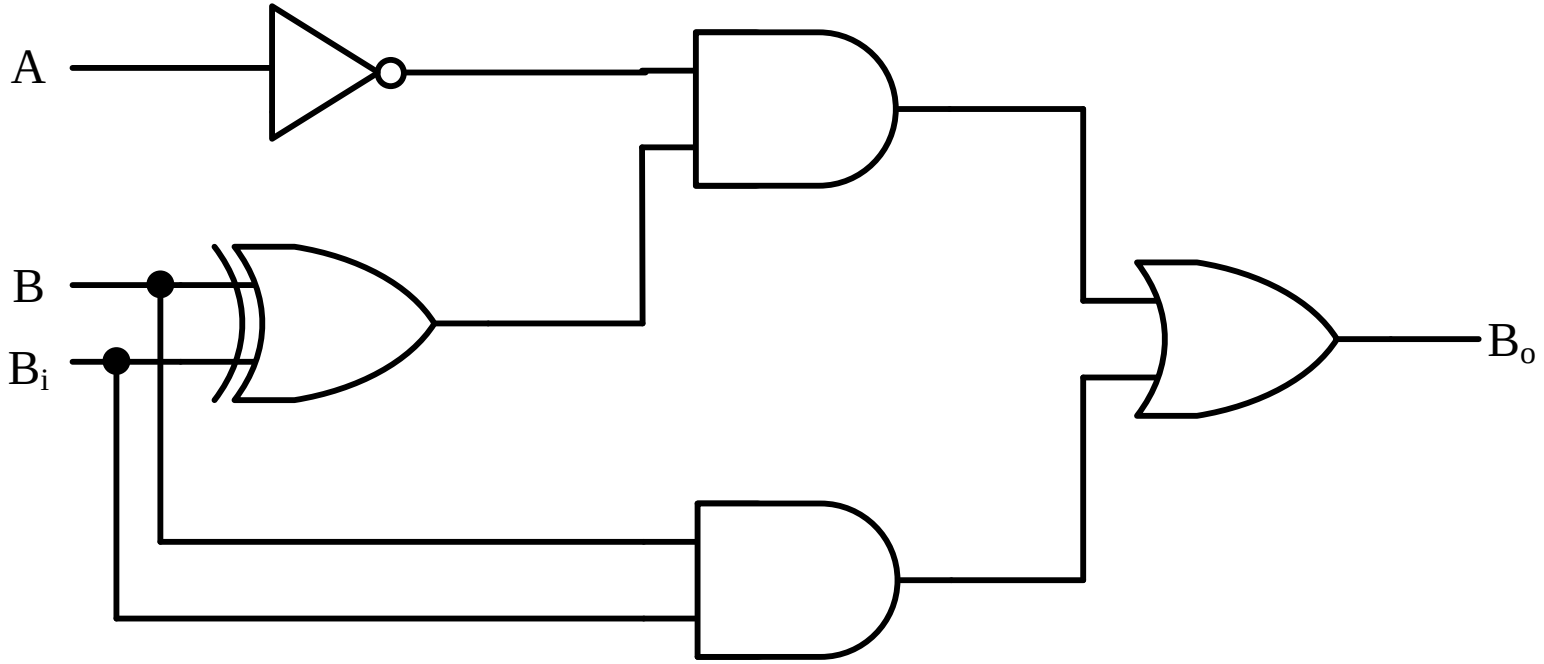
$$D = A^l (B^l B_i + B B_i^l) + A(B^l B_i^l + B B_i)$$

$$D = A^l (B \oplus B_i) + A((B \oplus B_i)^l)$$

$$\boxed{D = A \oplus B \oplus B_i}$$

FULL SUBTRACTOR CIRCUIT (OUTPUT BORROW)

A	B	B_i	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



$$B_o = A^l B^l B_i + A^l B B_i^l + A^l B B_i + A B B_i$$

$$B_o = A^l (B^l B_i + B B_i^l) + B B_i (A + A^l)$$

$$B_o = A^l (B \oplus B_i) + B B_i$$

SOLVED EXAMPLES

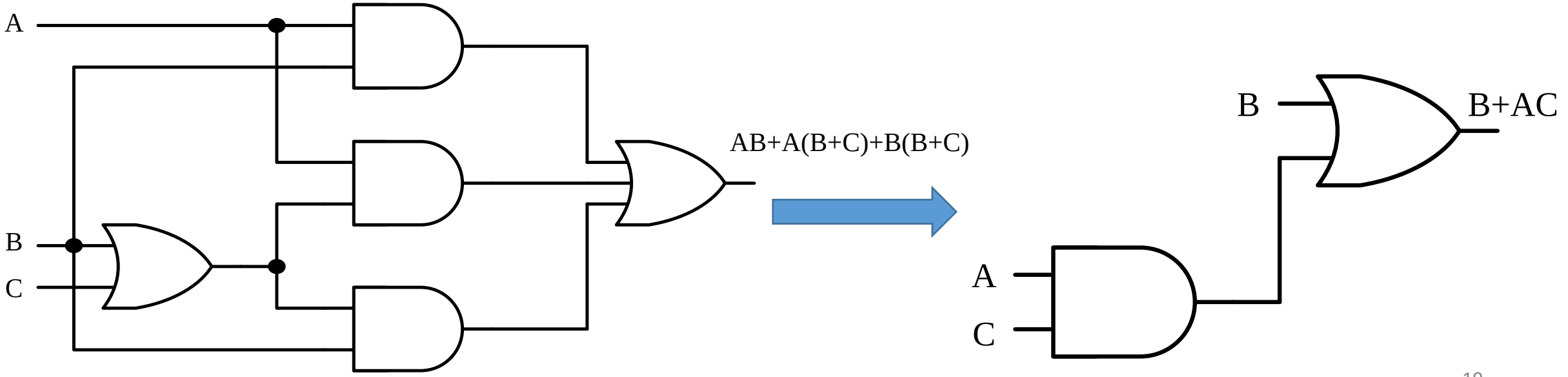
Q. Using Boolean algebra techniques, simplify this expression:

$$F = AB + A(B + C) + B(B + C)$$

Soln: $F = AB + A(B + C) + B(B + C) = AB + AB + AC + BB + BC$

$= AB + AC + B + BC = AC + BA + B.1 + BC = AC + B.(A + 1 + C)$

$= AC + B.1 = AC + B$



SOLVED EXAMPLES

Q. Minimize the following Boolean function using algebraic manipulation:

$$F=ABC^{l}D^{l}+ABC^{l}D+AB^{l}C^{l}D+ABCD+AB^{l}CD+ABCD^{l}+AB^{l}CD^{l}$$

Soln:

$$F = ABC^{l}(D+D^{l})+AB^{l}C^{l}D+ACD(B+B^{l})+ACD^{l}(B+B^{l})$$

$$F = ABC^{l} + AB^{l}C^{l}D + ACD + ACD^{l}$$

$$F = ABC^{l} + AB^{l}C^{l}D + AC(D + D^{l})$$

$$F = ABC^{l} + AB^{l}C^{l}D + AC$$

$$F = A(C+BC^{l}) + AB^{l}C^{l}D$$

$$F = A(C + B) + AB^{l}C^{l}D$$

$$F = AC + AB + AB^{l}C^{l}D$$

$$F = AC + A(B + B^{l}C^{l}D)$$

$$F = AC + A(B + C^{l}D) = AC + AB + AC^{l}D = A(C + C^{l}D) + AB$$

$$F = A(C + D) + AB = AC + AD + AB$$

SOLVED EXAMPLES

Q. Apply de Morgan's theorems to the following expressions:

(a) $F = \overline{\overline{\overline{A} + B\overline{C}} + D(E + \overline{F})}$ (b) $F = \overline{A\overline{B} + C\overline{D} + EF}$

Soln:

(a) $F = \overline{\overline{\overline{A} + B\overline{C}} + D(E + \overline{F})}$

$$F = \left(\overline{\overline{\overline{A} + B\overline{C}}}\right)\left(\overline{D(E + \overline{F})}\right) = \left(A + B\overline{C}\right)\left(\overline{D} + \overline{(E + \overline{F})}\right)$$

$$F = \left(A + B\overline{C}\right)\left(\overline{D} + E + \overline{F}\right)$$

(b) $F = \overline{A\overline{B} + C\overline{D} + EF}$

$$F = \left(\overline{A\overline{B}}\right)\left(\overline{C\overline{D}}\right)\left(\overline{EF}\right)$$

$$F = \left(\overline{A} + \overline{\overline{B}}\right)\left(\overline{C} + \overline{\overline{D}}\right)\left(\overline{E} + \overline{F}\right)$$

$$F = \left(\overline{A} + B\right)\left(\overline{C} + D\right)\left(\overline{E} + \overline{F}\right)$$

SOLVED EXAMPLES

Q. Draw the circuit to implement the function $F = \overline{\overline{A \cdot B} + \overline{A \cdot C}}$

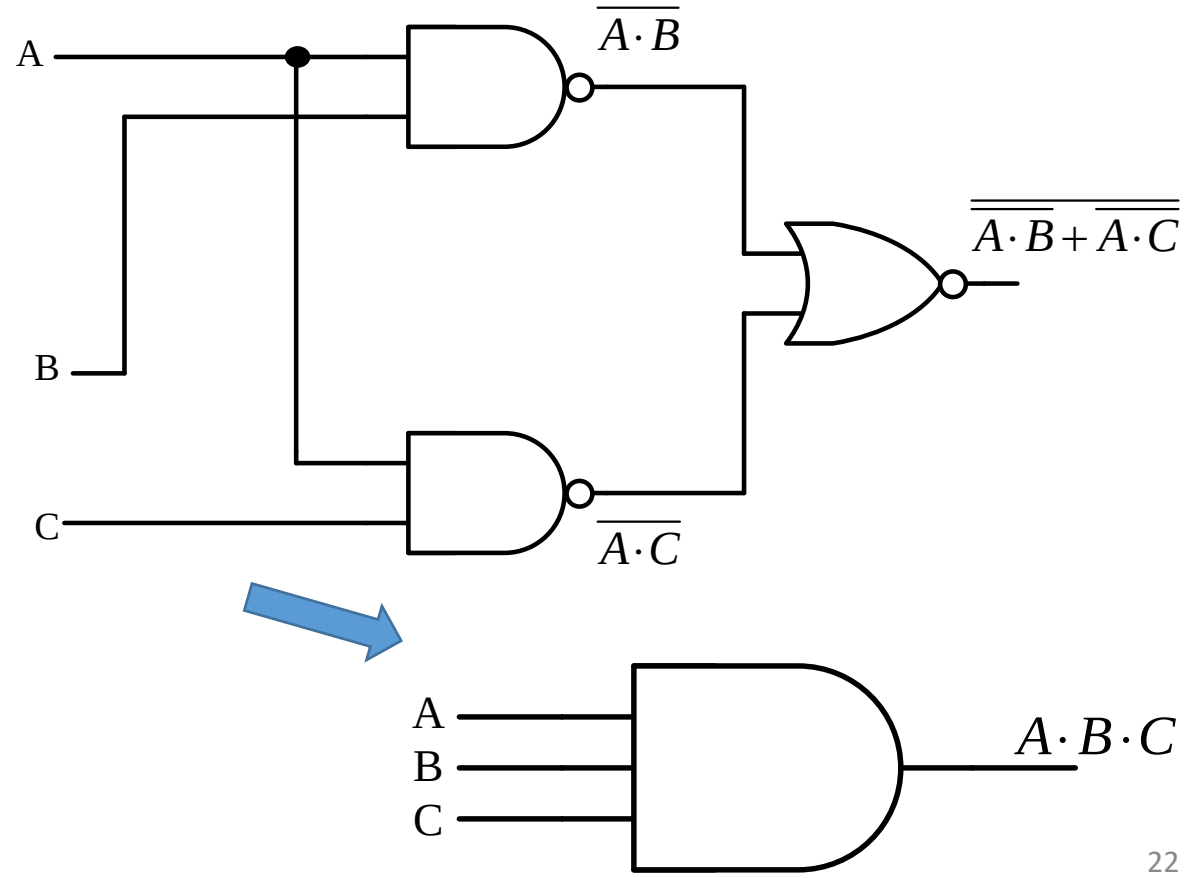
Simplify this function and hence redraw the reduced circuit.

Soln: $F = \overline{\overline{A \cdot B} + \overline{A \cdot C}}$

$$F = \left(\overline{\overline{A \cdot B}}\right)\left(\overline{\overline{A \cdot C}}\right)$$

$$F = A \cdot B \cdot A \cdot C$$

$$F = A \cdot B \cdot C$$



SUM OF PRODUCTS (SOP) FORM

- Two or more "product" (AND-gated) terms are "summed" (OR-gated) to form a Boolean expression.
- Examples: $AB + ABC$, $ABC + \bar{C}DE + \bar{B}C\bar{D}$, $AB + BCD + \bar{E}$, etc.
- In an SOP expression, a single overbar (denoting inversion) cannot extend over more than one variable e.g. there can be terms like $\bar{A}\bar{B}\bar{C}$, but terms like $\overline{ABC}$ are not allowed.
- In a "standard" SOP expression, each product term contains all the input variables e.g. for 3 inputs, $AB + A\bar{B}C$ is not in standard form, whereas $AB\bar{C} + A\bar{B}C$ is in standard form.

PRODUCT OF SUMS (POS) FORM

- Two or more "sum" (OR-gated) terms are taken "product" (AND-gated) to form a Boolean expression.
- Examples: $(\overline{A}+B)(A+\overline{B}+C)$, $(A+B+C)(\overline{B}+C+\overline{D})(\overline{C}+D)\overline{E}$, etc.
- In an POS expression, a single overbar (denoting inversion) cannot extend over more than one variable e.g. there can be terms like $\overline{A}+\overline{B}+\overline{C}$, but terms like $\overline{A+B+C}$ are not allowed.
- In a "standard" POS expression, each product term contains all the input variables e.g. for 3 inputs, $(A+B)(A+\overline{B}+C)$ is not in standard form, whereas $(A+B+\overline{C})(A+\overline{B}+C)$ is in standard form.

SOLVED EXAMPLES

Q. Convert the following Boolean expression into standard SOP form: $A\overline{B}C + \overline{A}\overline{B} + AB\overline{C}D$

The domain of this SOP expression A, B, C, D. Take one term at a time.

The first term, $A\overline{B}C$, is missing variable D or $\overline{D}$, so multiply the first term by (D+$\overline{D}$) as follows:

$$A\overline{B}C = A\overline{B}C(D + \overline{D}) = A\overline{B}CD + A\overline{B}C\overline{D}$$

The second term, $\overline{A}\overline{B}$, is missing variables C or $\overline{C}$ and D or $\overline{D}$, so first multiply the second term by (C+$\overline{C}$) as follows:

$$\overline{A}\overline{B} = \overline{A}\overline{B}(C + \overline{C}) = \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C}$$

The two resulting terms are missing variable D or $\overline{D}$, so multiply both terms by (D+$\overline{D}$) as follows:

$$\overline{A}\overline{B}C(D + \overline{D}) + \overline{A}\overline{B}\overline{C}(D + \overline{D}) = \overline{A}\overline{B}CD + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D}$$

The third term, AB$\overline{C}$D, is already in standard form.

The complete standard SOP form of the original expression is as follows:

$$A\overline{B}CD + A\overline{B}C\overline{D} + \overline{A}\overline{B}CD + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D} + AB\overline{C}D$$

SOLVED EXAMPLES

Q. Convert the following Boolean expression into standard POS form: $(\overline{A}+B+C)(\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$

The domain of this POS expression A, B, C, D. Take one term at a time.

The first term, $\overline{A}+B+C$, is missing variable D or $\overline{D}$, so add $D\overline{D}$ as follows:

$$\overline{A}+B+C=\overline{A}+B+C+D\overline{D}=(\overline{A}+B+C+D)(\overline{A}+B+C+\overline{D})$$

Note, $A+BC=(A+B)(A+C)$

The second term, $\overline{B}+C+\overline{D}$, is missing variables A or $\overline{A}$ and D or $\overline{D}$, so $A\overline{A}$ as follows:

$$\overline{B}+C+\overline{D}=\overline{B}+C+\overline{D}+A\overline{A}=(\overline{B}+C+\overline{D}+A)(\overline{B}+C+\overline{D}+\overline{A})=(A+\overline{B}+C+\overline{D})(\overline{A}+\overline{B}+C+\overline{D})$$

The third term, $A+\overline{B}+\overline{C}+D$, is already in standard form.

The complete standard POS form of the original expression is as follows:

$$(\overline{A}+B+C)(\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)=$$

$$(\overline{A}+B+C+D)(\overline{A}+B+C+\overline{D})(A+\overline{B}+C+\overline{D})(\overline{A}+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$$

EXERCISES

Q. Convert the following SOP expression into standard SOP form:

a. $AB + \overline{A}BD + \overline{A}C\overline{D}$

b. $\overline{A}BC + AB\overline{C}$

Q. Convert the following POS expression into standard POS form:

a. $\left(A + \overline{B} + C\right)\left(A + B + \overline{C}\right)$

b. $A\left(A + \overline{C}\right)(A + B)$

Canonical Forms of Boolean Expressions

- The various possible standard product (AND) terms are called "minterms", while the standard sum (OR) terms are called "maxterms".
- N input variables (bits) can be combined to form 2^N minterms or 2^N maxterms, designated by their decimal equivalents (bits are assigned values such that each minterm yields value 1 while each maxterm yields value 0)

X	Y	Z	minterm	designation	maxterm	designation
0	0	0	$\bar{X}\bar{Y}\bar{Z}$	m_o	$X+Y+Z$	M_o
0	0	1	$\bar{X}\bar{Y}Z$	m_1	$X+Y+\bar{Z}$	M_1
0	1	0	$\bar{X}Y\bar{Z}$	m_2	$X+\bar{Y}+Z$	M_2
0	1	1	$\bar{X}YZ$	m_3	$X+\bar{Y}+\bar{Z}$	M_3
1	0	0	$X\bar{Y}\bar{Z}$	m_4	$\bar{X}+Y+Z$	M_4
1	0	1	$X\bar{Y}Z$	m_5	$\bar{X}+Y+\bar{Z}$	M_5
1	1	0	$XY\bar{Z}$	m_6	$\bar{X}+\bar{Y}+Z$	M_6
1	1	1	XYZ	m_7	$\bar{X}+\bar{Y}+\bar{Z}$	M_7

Note that each maxterm is the complement (inverse) of the corresponding minterm

i.e. $\boldsymbol{m_i = M_i'}$

(from de Morgan's laws)

Canonical Forms of Boolean Expressions

Example: We can write the expression for function *F* using the truth table below

X	Y	Z	F
0	0	0	**0**
0	0	1	**1**
0	1	0	**0**
0	1	1	**0**
1	0	0	**1**
1	0	1	**0**
1	1	0	**0**
1	1	1	**1**

$$F = \overline{X}\,\overline{Y}Z + X\overline{Y}\,\overline{Z} + XYZ$$

$$F = m_1 + m_4 + m_7$$

(**SOP** form using **minterms**: collect all the terms which give ***F* = 1**)

$$F = (X + Y + Z)(X + \overline{Y} + Z)(X + \overline{Y} + \overline{Z})(\overline{X} + Y + \overline{Z})(\overline{X} + \overline{Y} + Z)$$

$$F = M_0 M_2 M_3 M_5 M_6$$

(**POS** form using **maxterms**: collect all the terms which give ***F* = 0**)

Any Boolean function can be expressed as a sum of minterms (SOP form) or product of maxterms (POS form).

SOLVED EXAMPLES

Q. Express the Boolean function $F = A + \overline{B}C$ as a sum of minterms (SOP).

First term missing B and C variables,

$$A = A(B+\overline{B})(C+\overline{C}) = \left(AB + A\overline{B}\right)\left(C+\overline{C}\right) = ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C}$$

Second term missing variable A,

$$\overline{B}C = \overline{B}C(A+\overline{A}) = A\overline{B}C + \overline{A}\,\overline{B}C$$

$$F = ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C} + A\overline{B}C + \overline{A}\,\overline{B}C = ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}\,\overline{B}C$$

$$F = m_7 + m_6 + m_5 + m_4 + m_1$$

In short notation

$$F(A,B,C) = \sum(1,4,5,6,7)$$

$$\overline{F}(A,B,C) = \sum(0,2,3)$$

SOLVED EXAMPLES

Q. Express $F = XY + \overline{X}Z$ as a product of maxterms (POS form).

$F = XY + \overline{X}Z$

$F = \left(XY + \overline{X}\right)(XY + Z) = \left(X + \overline{X}\right)\left(Y + \overline{X}\right)(X + Z)(Y + Z)$

$F = \left(Y + \overline{X}\right)(X + Z)(Y + Z) \quad \left(\therefore X + \overline{X} = 1\right)$

$F = \left(\overline{X} + Y + Z\overline{Z}\right)\left(X + Y\overline{Y} + Z\right)\left(X\overline{X} + Y + Z\right) \quad (\therefore X\overline{X} = 0)$

$F = \left(\overline{X} + Y + Z\right)\left(\overline{X} + Y + \overline{Z}\right)(X + Y + Z)\left(X + \overline{Y} + Z\right)(X + Y + Z)\left(\overline{X} + Y + Z\right) \quad \left(\therefore X + Y\overline{Y} = (X + Y)(X + \overline{Y})\right)$

$F = \left(\overline{X} + Y + Z\right)\left(\overline{X} + Y + \overline{Z}\right)(X + Y + Z)\left(X + \overline{Y} + Z\right) \quad (\therefore XX = X)$

$F = M_4 M_5 M_0 M_2$

$F(X, Y, Z) = \Pi(0, 2, 4, 5)$

$\overline{F}(X, Y, Z) = \Pi(1, 3, 6, 7)$

EXERCISES

Q. Express the Boolean function $F = X + YZ$ as a sum of minterms.

Q. Express the Boolean function $F = X + YZ$ as a product of maxterms.

KARNAUGH MAP (K-MAP)

- Karnaugh maps are a powerful, widely used, **tabular method of logic simplification** that are used to find the simplest SOP or POS form.

- Karnaugh maps (or just **K-maps**) allow expressions with up to four variables to be handled in a straightforward manner, and can be used with expressions of **up to six Boolean variables**.

- **Each cell** in the Karnaugh map corresponds to **one line** in the **truth table**.

KARNAUGH MAP (K-MAP)

Steps to simplify a logic expression using a Karnaugh map are:

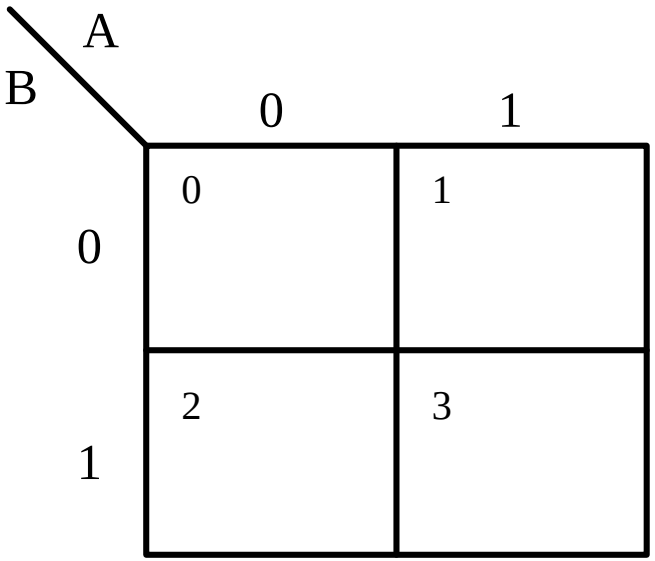
1. Draw the empty Karnaugh map.
2. Fill in 1's and 0's into the corresponding cells. (using the original logic expression or it's truth table.)
3. Identify and draw loops on the Karnaugh map to group together adjacent 1's (for SOP form) or adjacent 0's (for POS form).
4. Create the expression for each group and combine to give the simplified logic expression.

KARNAUGH MAP (K-MAP)

1. Creating an empty Karnaugh map

- The **number of inputs N** defines the **number of cells (2^N)** in the Karnaugh map to be drawn.
- e.g. for two variables, A and B, there are four possible input states i.e. 00, 01, 11 and 10. Hence, the K-map will have 4 cells, **each cell corresponding to one particular input state**.
- The small numbers in the top left corner of **each cell** show the **decimal equivalent** of the binary input code for that cell. Notice that the cells are arranged such that **only 1-bit changes** when we go to an **adjacent cell**.
- Each cell thus corresponds to one particular **minterm m_i** or **maxterm M_i**, where ***i*** is the **decimal equivalent** denoting the **cell**.

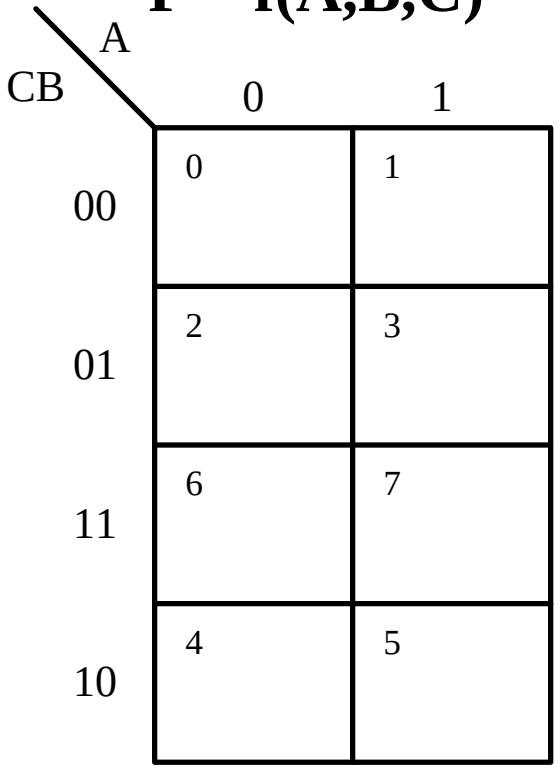
Two-variable K-map
F = f(A,B)

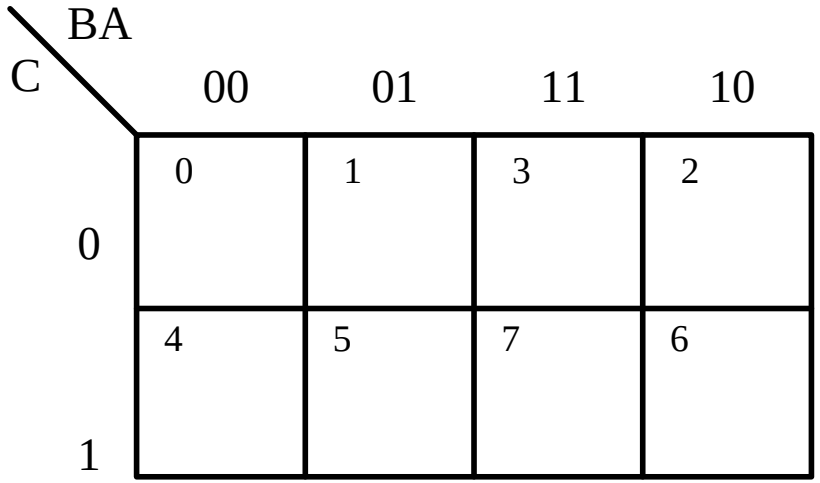


Truth table for two variables

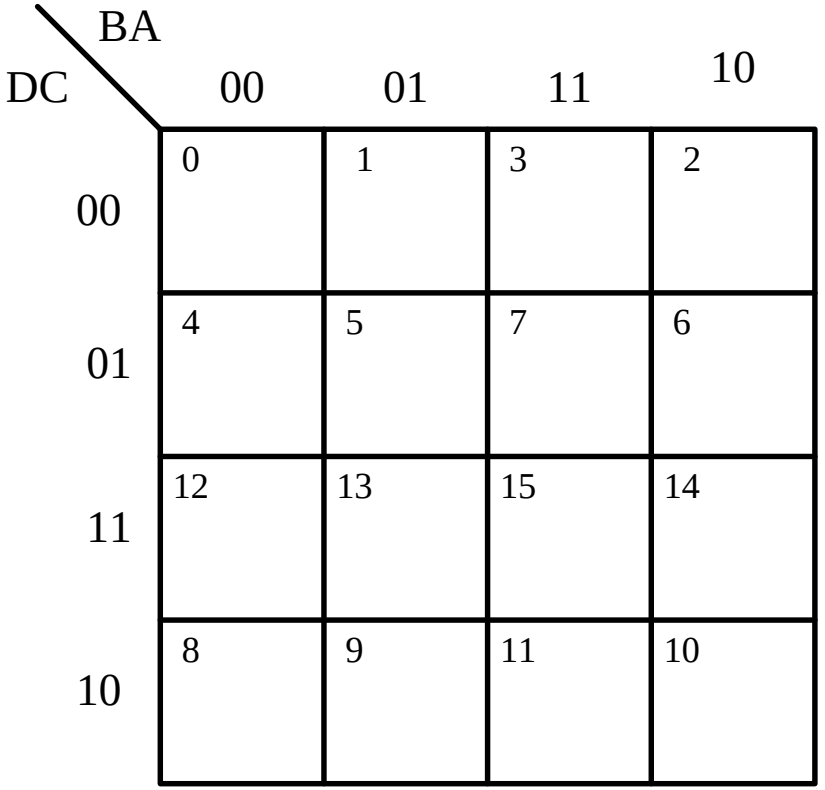
B	A	Output F	Binary code/ Decimal eq.
0	0		$00_2 = 0_{10}$
0	1		$01_2 = 1_{10}$
1	0		$10_2 = 2_{10}$
1	1		$11_2 = 3_{10}$

Three-variable K-map
F = f(A,B,C)





Four-variable K-map
F = f(A,B,C,D)



KARNAUGH MAP

2. Filling in the Karnaugh map

- **Each cell** of the Karnaugh map is filled with **either a 1 or a 0** signifying the **output** for the input corresponding to that cell.

- The **original binary function** which needs to be simplified may be **provided** in any one of the following forms:

- by specifying the binary/decimal numbers that correspond to an output of 1 (i.e. **sum of minterms**) or output of 0 (i.e. **product of maxterms**);
- a **general logical/Boolean expression**;
- a **truth table**.

K-MAP FILLING EXAMPLE

Q. Map the following standard SOP expression on a Karnaugh map:

$$\bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + AB\bar{C}D + ABCD + AB\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

Given expression is a sum of minterms:

$$F = \sum(0011, 0100, 1101, 1111, 1100, 0001, 1010)$$

$$= \sum m(3, 4, 13, 15, 12, 1, 10)$$

Fill in 1's into the cells corresponding to the above inputs
(rest of the cells have output 0)

AB \ CD	00	01	11	10
00		1	1	
01	1			
11	1	1	1	
10				1

K-MAP FILLING EXAMPLE

Q. Map the following SOP expression on a Karnaugh map: $\overline{A} + A\overline{B} + AB\overline{C}$

The SOP expression is not in standard form because each product term does not have three variables.

The first term is missing two variables, the second term is missing one variable, and the third term is standard.

First expand the terms numerically as follows,

$$\overline{A} = \overline{A}(B+\overline{B})(C+\overline{C}) = \left(\overline{A}B + \overline{A}\,\overline{B}\right)\left(C+\overline{C}\right) = \overline{A}BC + \overline{A}B\overline{C} + \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C}$$

The second term is expand as,

$$A\overline{B} = A\overline{B}(C+\overline{C}) = A\overline{B}C + A\overline{B}\overline{C}$$

The standard SOP expression is given as,

$$\overline{A}BC + \overline{A}B\overline{C} + \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C + A\overline{B}\overline{C} + AB\overline{C}$$

$$F = \sum(011, 010, 001, 000, 101, 100, 110)$$
$$= \sum m(3, 2, 1, 0, 5, 4, 6)$$

AB \ C	0	1
00	1	1
01	1	1
11	1	
10	1	1

K-MAP FILLING EXAMPLE

Q. Map the following standard POS expression on a Karnaugh map:

$$(\overline{A}+\overline{B}+C+D)(\overline{A}+B+\overline{C}+\overline{D})(A+B+\overline{C}+D)(\overline{A}+\overline{B}+\overline{C}+\overline{D})(A+B+\overline{C}+\overline{D})$$

Given expression is a product of maxterms:

$F = \prod(1100, 1011, 0010, 1111, 0011)$

$= \prod M(12, 13, 2, 15, 3)$

Fill in 0's into the cells corresponding to the above inputs
(rest of the cells have output 1)

AB \ CD	00	01	11	10
00			0	0
01				
11	0		0	
10			0	

K-MAP FILLING EXAMPLE

Q. Map the following standard POS expression on a Karnaugh map:

$$(A+B+C)(A+\overline{B}+C)(\overline{A}+\overline{B}+C)(\overline{A}+B+\overline{C})$$

Given expression is a product of maxterms:

$F = \prod(000, 010, 110, 101)$

$= \prod M(0, 2, 6, 5)$

Fill in 0's into the cells corresponding to the above inputs
(rest of the cells have output 1)

AB \ C	0	1
00	0	
01	0	
11	0	
10		0

K-MAP FILLING EXERCISES

Q. Map the following SOP expression on a Karnaugh map:

$$\overline{B}\overline{C} + A\overline{B} + AB\overline{C} + A\overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}D + A\overline{B}CD$$

Q. Map the following POS expression on a Karnaugh map

$$\left(B + C + D\right)\left(A + B + \overline{C} + D\right)\left(\overline{A} + B + C + \overline{D}\right)\left(A + \overline{B} + C + D\right)\left(\overline{A} + \overline{B} + C + D\right)$$

KARNAUGH MAP

3. Creating groups in Karnaugh maps

When all the output values are entered, all of the 1's (or all of the 0's, depending on if SOP or POS form is finally required, respectively) must be **grouped together** according to the following rules:

- Each **group** must contain **2^n adjacent cells**, where n is an +ve integer.
- Each **group** should be made **as large as possible**, while satisfying the constraint of having cells numbering to a power of 2.
- The **larger the groups** formed, the **simpler the final function**.

KARNAUGH MAP

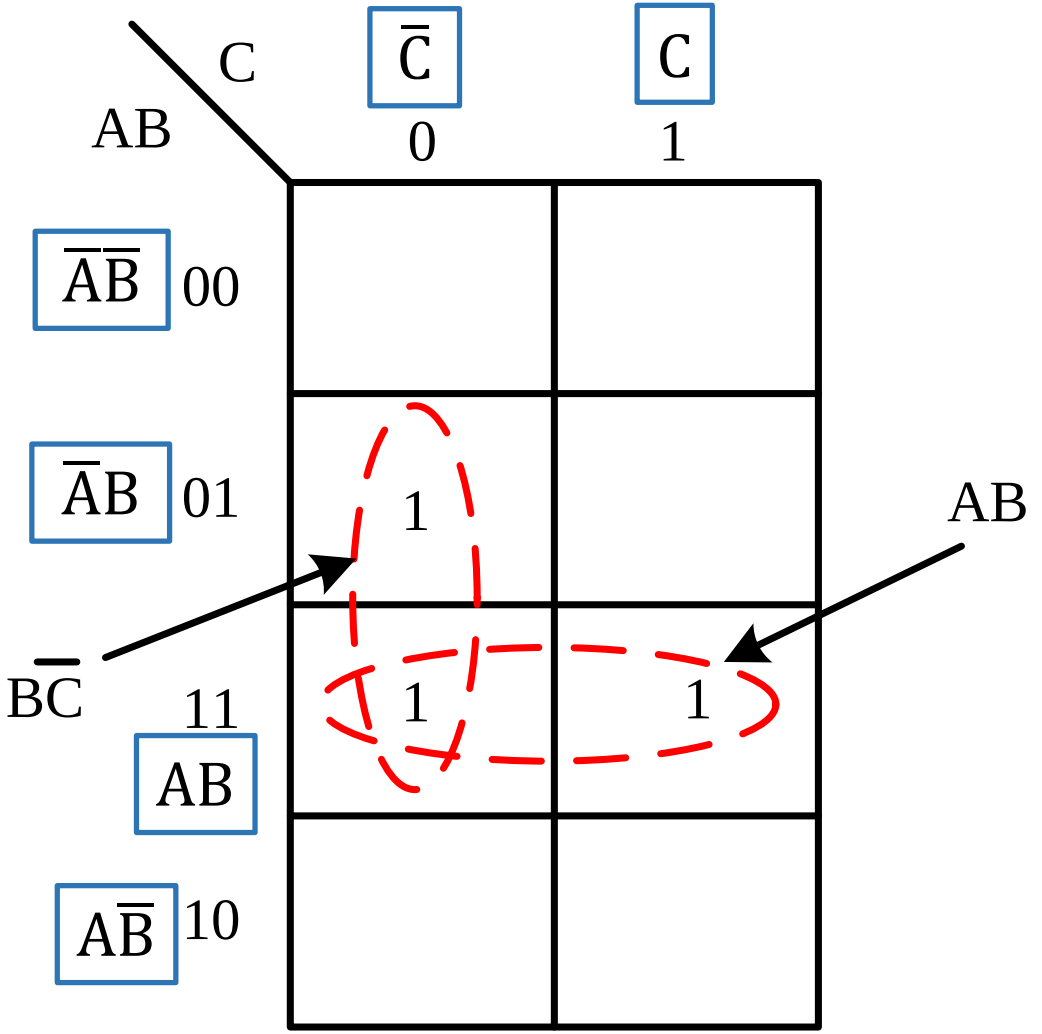
3. Creating groups in Karnaugh maps

- **Each cell** with output **1** (or each cell with output **0**; for **SOP** or **POS**) must be included in **at least one group**.
- The **groups may overlap**, so one cell may be included in several groups, but any group that has all its elements included in other groups can be ignored.
- There may be **several correct minimal forms** for a given logic function, **dependent upon the particular groupings** that are chosen.
- Once the groups have been assigned, a logic expression can be created. **For each group**, write down the **common inputs** (AND/product terms if using 1s; OR/sum terms if using 0s) that describe that specific group and combine accordingly (SOP or POS).

SOLVED EXAMPLES

Q. Simplify $F = \overline{A}B\overline{C} + AB\overline{C} + ABC$ into simpler SOP form using K-map

$$F = AB + B\overline{C}$$



SOLVED EXAMPLES

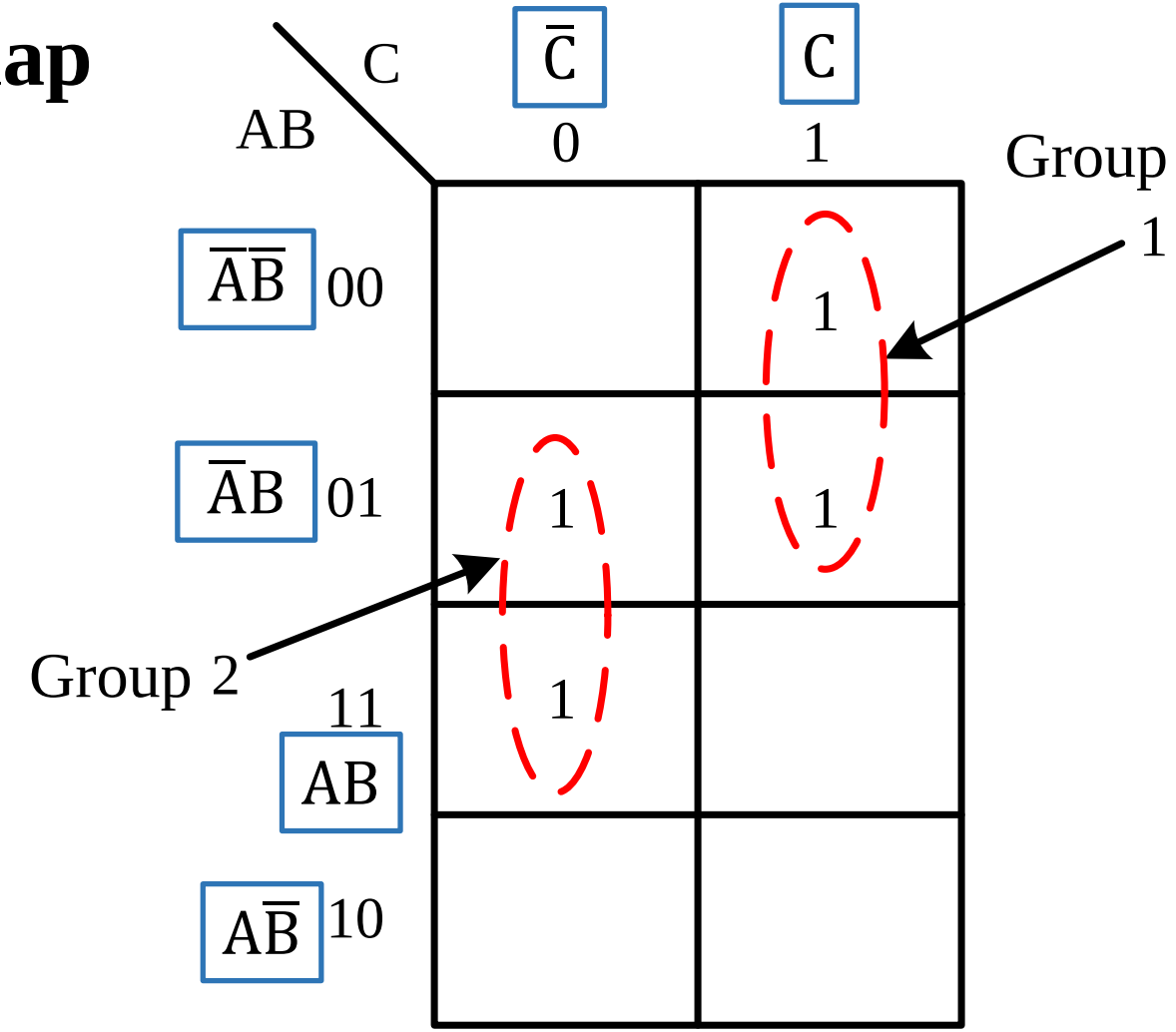
Q. Simplify $F = \sum m(1,2,3,6)$

into simpler SOP form using K-map

$$\text{Group } 1 \Rightarrow \overline{A}C$$

$$\text{Group } 2 \Rightarrow B\overline{C}$$

$$F = \overline{A}C + B\overline{C}$$



SOLVED EXAMPLES

Q. Simplify $F = \sum m(2,3,4,5,7,8,10,13,15)$

into simpler SOP form using K-map

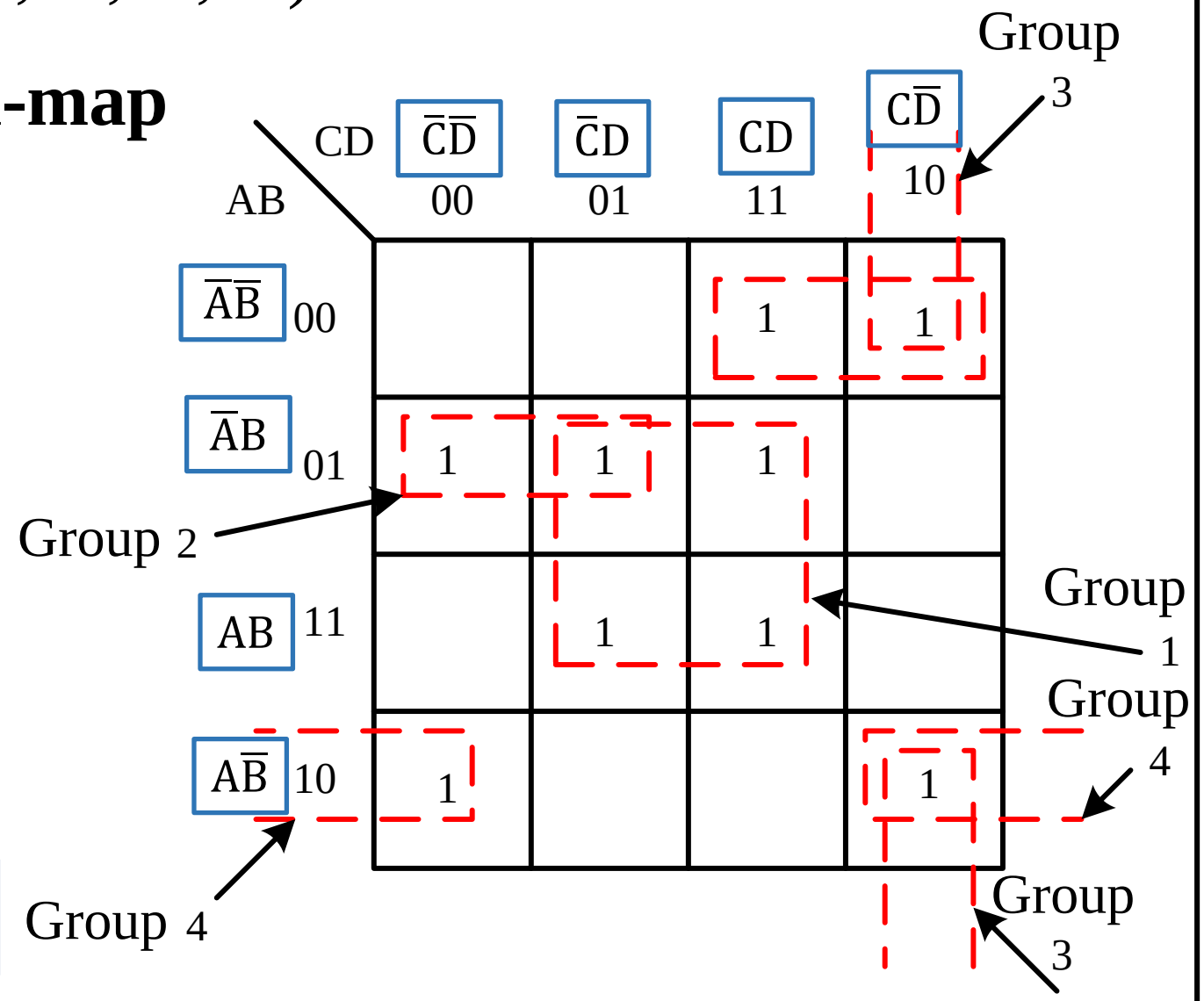
Group

$1 \Rightarrow BD$

$2 \Rightarrow \overline{A}B\overline{C}$

$3 \Rightarrow \overline{B}C\overline{D}$

$4 \Rightarrow A\overline{B}\overline{D}$



$$F = BD + \overline{A}B\overline{C} + \overline{B}C\overline{D} + A\overline{B}\overline{D}$$

SOLVED EXAMPLES

Q. Simplify $F = \sum m(0,5,7,8,10,12,14,15)$

into simpler SOP form using K-map

Group

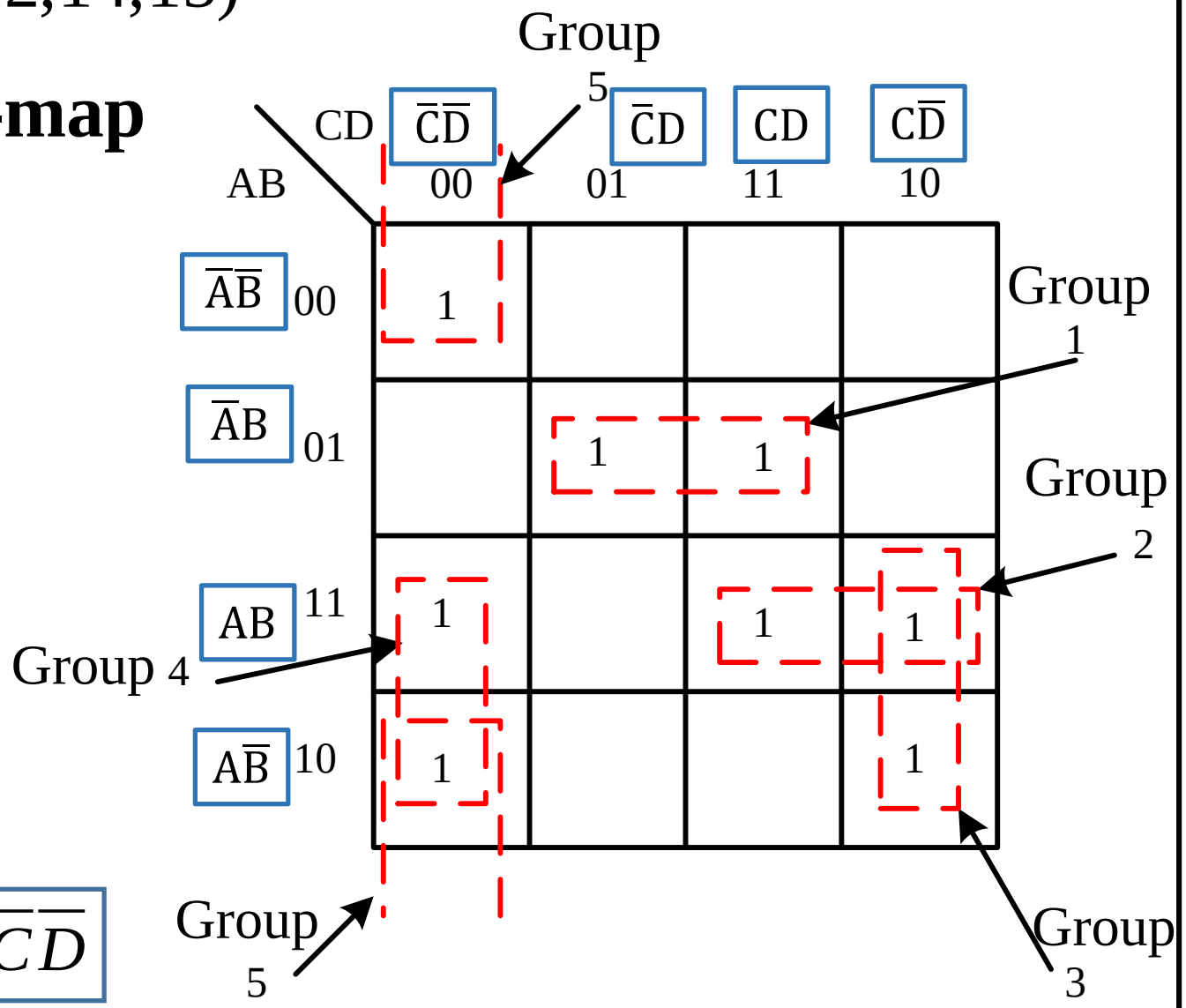
$1 \Rightarrow \overline{A}BD$

$2 \Rightarrow ABC$

$3 \Rightarrow AC\overline{D}$

$4 \Rightarrow A\overline{C}\,\overline{D}$

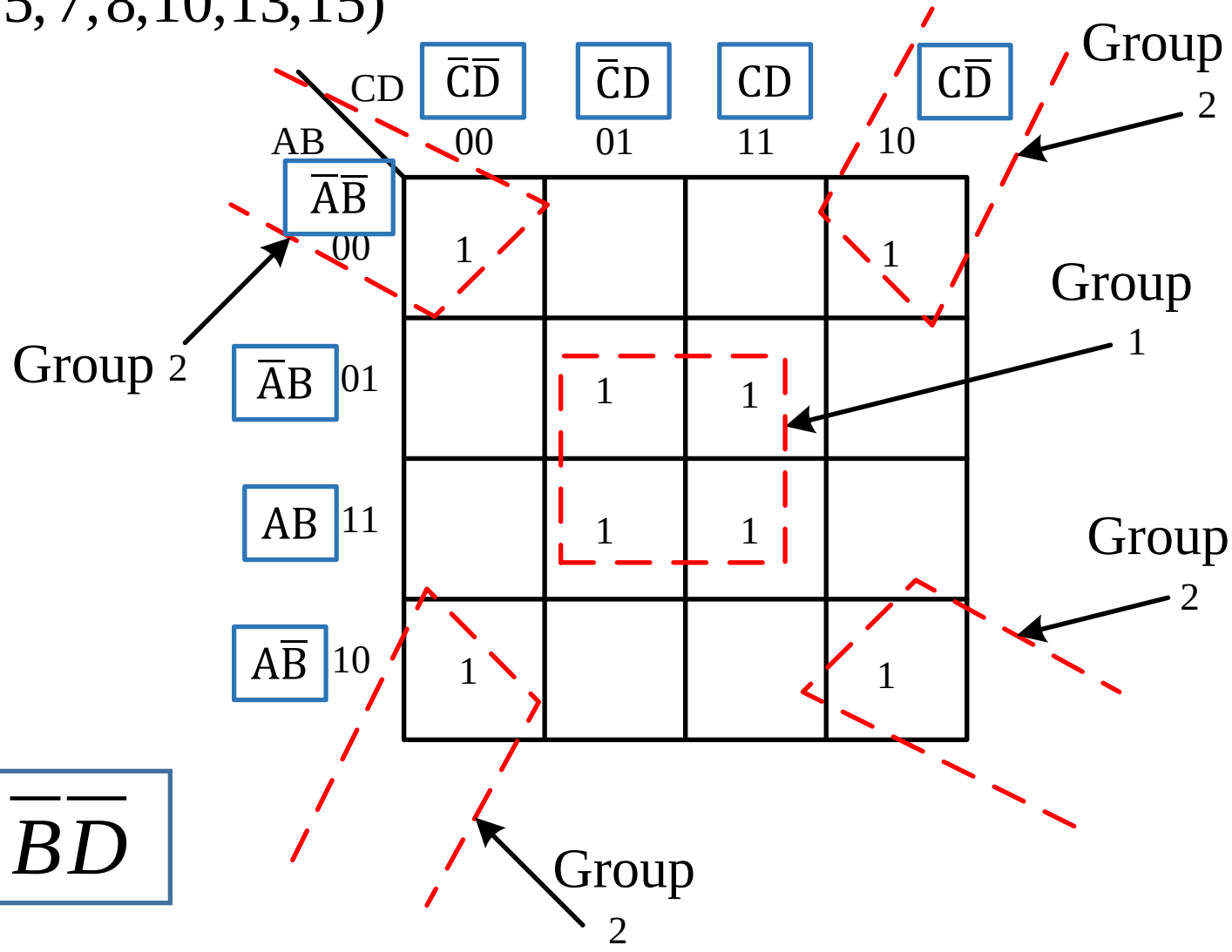
$4 \Rightarrow \overline{B}\,\overline{C}\,\overline{D}$



$$F = \overline{A}BD + ABC + AC\overline{D} + A\overline{C}\,\overline{D} + \overline{B}\,\overline{C}\,\overline{D}$$

SOLVED EXAMPLES

Q. Use a Karnaugh map to find the minimum SOP expression of the function: $f(A,B,C,D) = \sum m(0,2,5,7,8,10,13,15)$



Group

$1 \Rightarrow BD$

$2 \Rightarrow \bar{B}\bar{D}$

Minimum SOP: $F = BD + \bar{B}\bar{D}$

SOLVED EXAMPLES

Q. Use a Karnaugh map to minimize the following standard POS expression:

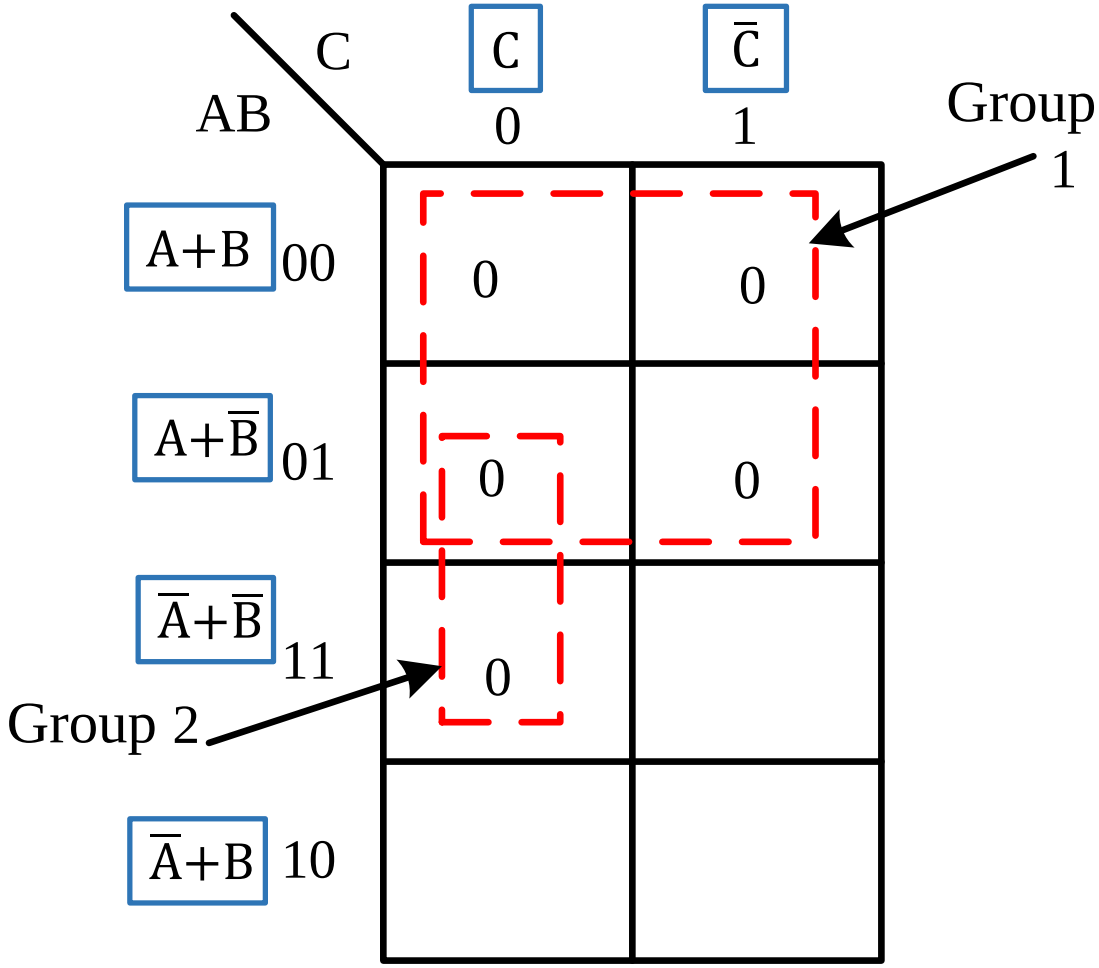
$$F = (A+B+C)(A+B+\overline{C})(A+\overline{B}+C)(A+\overline{B}+\overline{C})(\overline{A}+\overline{B}+C)$$

Group

$1 \Rightarrow A$

$2 \Rightarrow (\overline{B}+C)$

$$\boxed{F = A(\overline{B}+C)}$$



SOLVED EXAMPLES

Q. Use K-maps to convert the following standard POS expression into:
a) a minimum POS expression, b) a standard SOP expression, and
c) a minimum SOP expression.

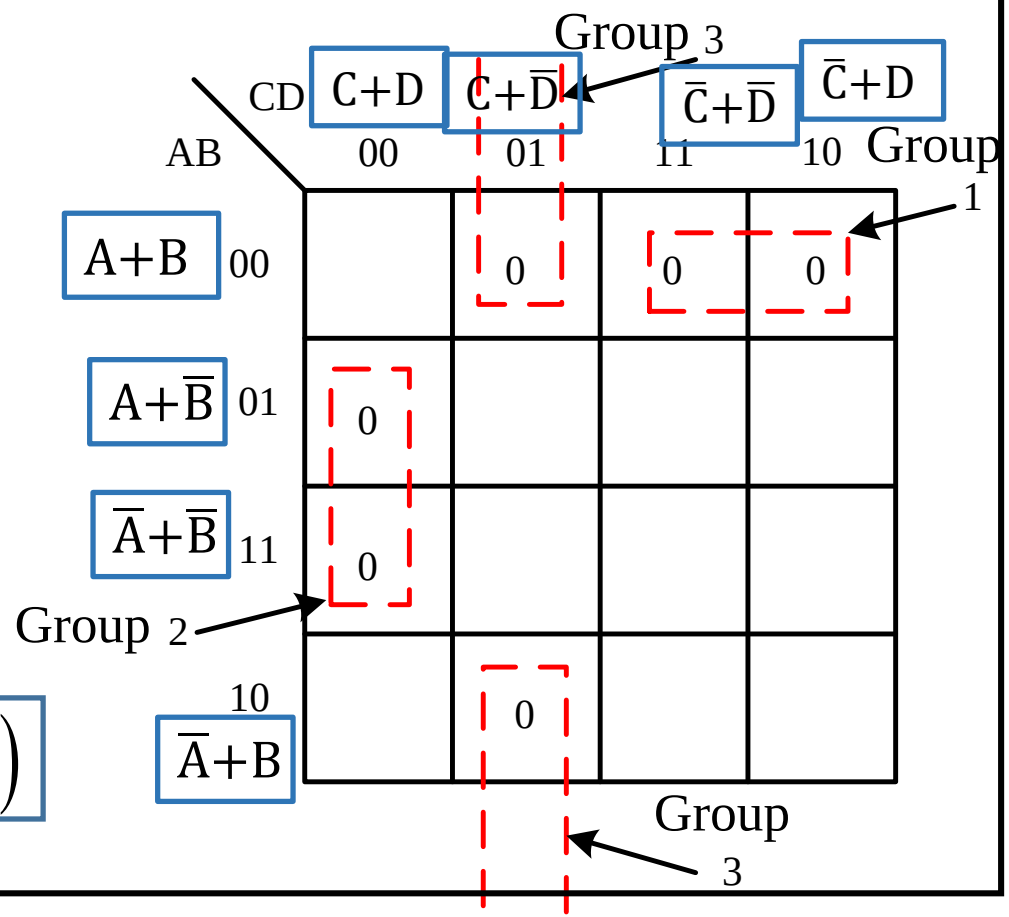
$$F = (\overline{A}+\overline{B}+C+D)(A+\overline{B}+C+D)(A+B+C+\overline{D})(A+B+\overline{C}+\overline{D})(\overline{A}+B+C+\overline{D})(A+B+\overline{C}+D)$$

a)

Group
$1 \Rightarrow \left(A+B+\overline{C}\right)$

$2 \Rightarrow \left(\overline{B}+C+D\right)$

$3 \Rightarrow \left(B+C+\overline{D}\right)$



Minimum POS: $F = \left(A+B+\overline{C}\right)\left(\overline{B}+C+D\right)\left(B+C+\overline{D}\right)$

b) The remaining cells are filled with 1s.

Sum all the minterms corresponding to these cells with output 1 to get the **standard SOP** expression as:

$$F = \sum(0000, 0101, 0111, 0110, 1101, 1111, 1110, 1000, 1011, 1010)$$

$$F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D} + AB\bar{C}D + ABCD + ABC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D}$$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1	0	0	0
$\bar{A}B$ 01	0	1	1	1
AB 11	0	1	1	1
$A\bar{B}$ 10	1	0	1	1

c)

Group

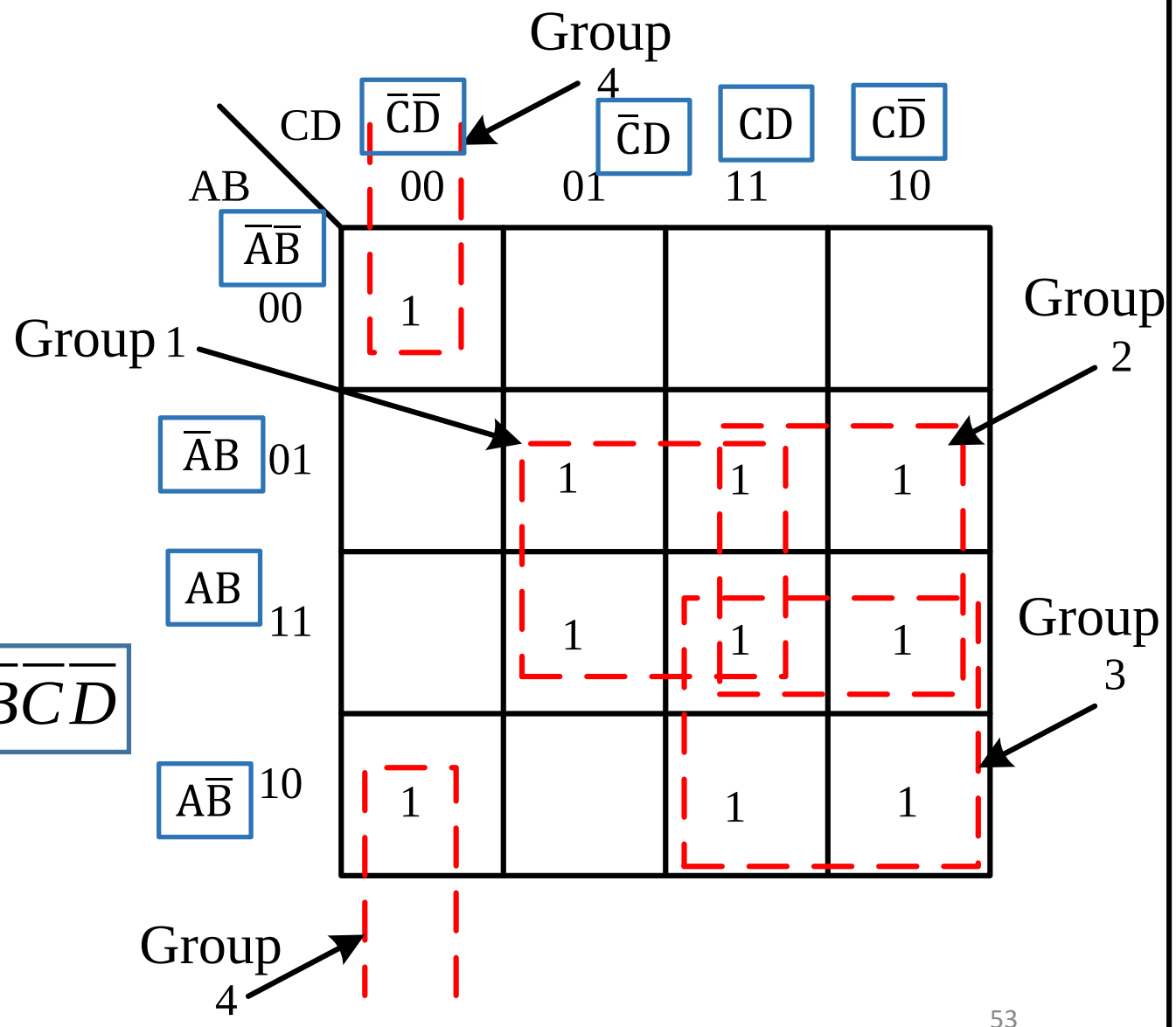
$1 \Rightarrow BD$

$2 \Rightarrow BC$

$3 \Rightarrow AC$

$4 \Rightarrow \overline{B}\overline{C}\overline{D}$

Minimum SOP: $F = BD + BC + AC + \overline{B}\overline{C}\overline{D}$



KARNAUGH MAP

"Don't Care" Conditions

- Sometimes a situation arises in which **some input variable combinations are not allowed**. For example, in a particular coding scheme with 4 input bits called “BCD”, there are six invalid combinations: 1010, 1011, 1100, 1101, 1110, and 1111.

- Since these unallowed states will never occur in for that particular application, they can be treated as **"don't care" terms** with respect to their effect on the output. That is, for these "don't care" terms **either a 1 or a 0 may be assigned to the output**: it really **does not matter** since they will never occur.

KARNAUGH MAP

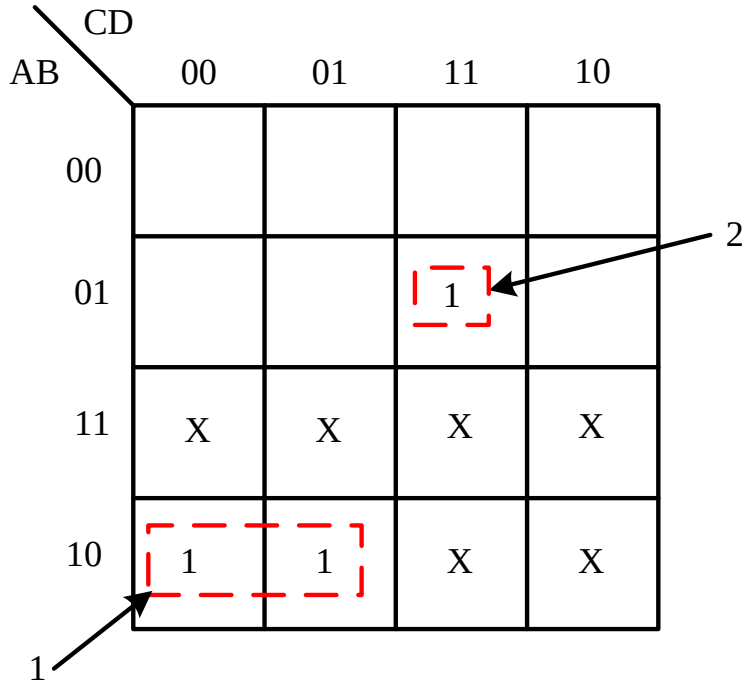
"Don't Care" Conditions

- The **"don't care" terms** can be used to our advantage on the Karnaugh map to **further simplify** the **Boolean expression** for a particular application.
- For **each "don't care" term**, an **X** is placed in the cell. When grouping the 1s, some of the **Xs can be treated as 1s** to make a larger grouping, while others **can be treated as 0s** if they cannot be used (and vice-versa if we are grouping 0s).
- The **larger** a **group**, the **simpler** the resulting **term** will be.

"Don't Care" Conditions

- Example: SOP form

Inputs ABCD	Output Y
0000	0
0001	0
0010	0
0011	0
0100	0
0101	0
0110	0
0111	1
1000	1
1001	1
1010	X
1011	X
1100	X
1101	X
1110	X
1111	X

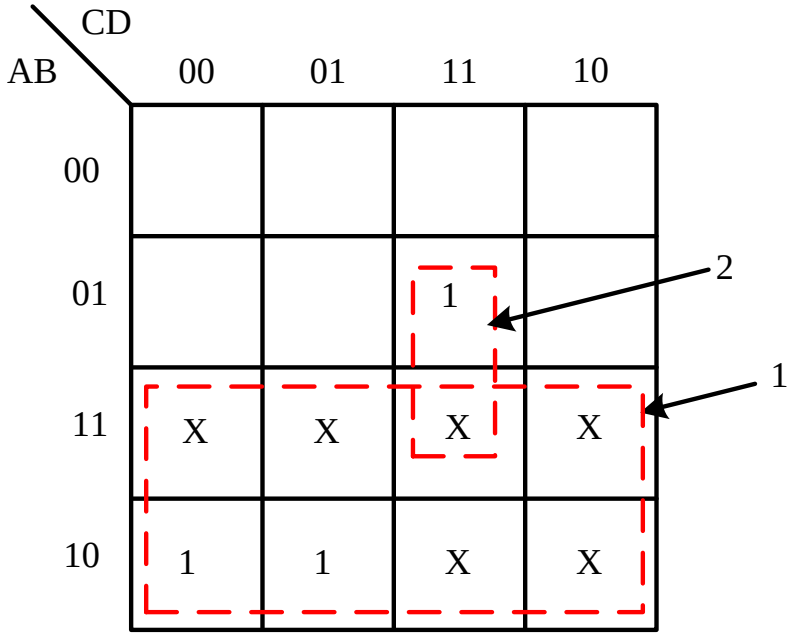


Without "don't cares"

$1 \Rightarrow A\overline{B}\,\overline{C}$

$2 \Rightarrow \overline{A}BCD$

$Y = A\overline{B}\,\overline{C} + \overline{A}BCD$



With "don't cares"

$1 \Rightarrow A$

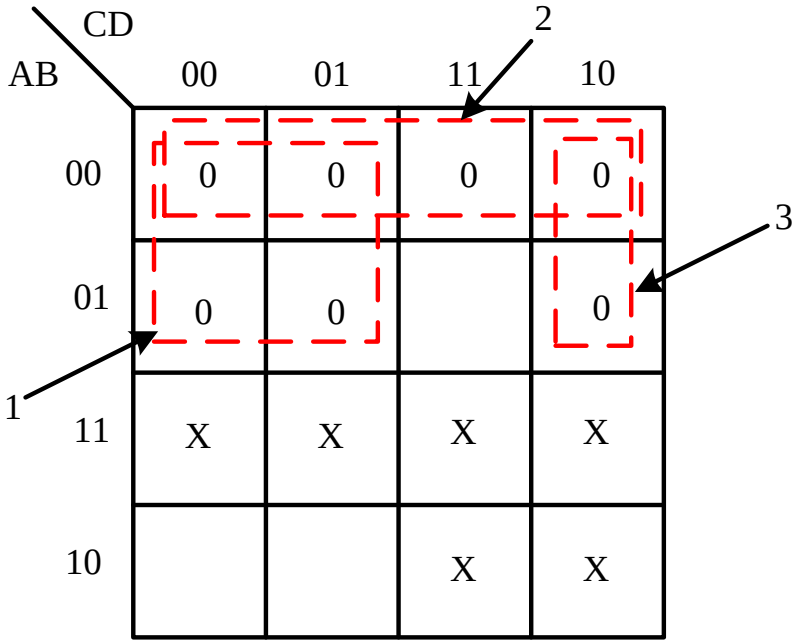
$2 \Rightarrow BCD$

$Y = A + BCD$

"Don't Care" Conditions

- Example: POS form

Inputs ABCD	Output Y
0000	0
0001	0
0010	0
0011	0
0100	0
0101	0
0110	0
0111	1
1000	1
1001	1
1010	X
1011	X
1100	X
1101	X
1110	X
1111	X



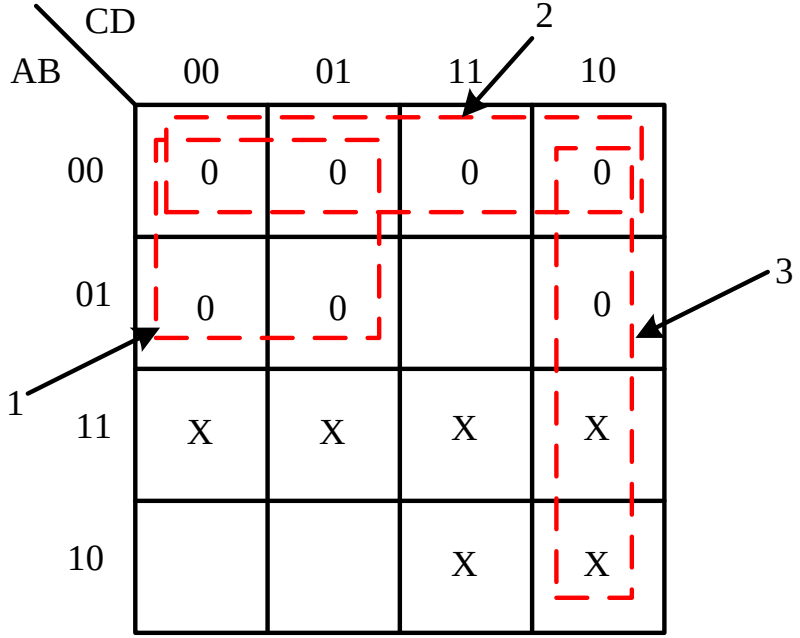
Without "don't cares"

$1 \Rightarrow A + C$

$2 \Rightarrow A + B$

$3 \Rightarrow A + \overline{C} + D$

$$Y = (A + C)(A + B)(A + \overline{C} + D)$$



With "don't cares"

$1 \Rightarrow A + C$

$2 \Rightarrow A + B$

$3 \Rightarrow \overline{C} + D$

$$Y = (A + C)(A + B)(\overline{C} + D)$$

EXERCISES

Q. Use Karnaugh maps to find the minimum SOP and POS expressions for the function $f(x_1,....x_4) = \overline{x}_1\overline{x}_3\overline{x}_4 + x_3x_4 + \overline{x}_1\overline{x}_2x_4 + x_1x_2\overline{x}_3x_4$, assuming that there are also don't-cares defined as $D = \sum(9,12,14)$

Q. Using K-maps, determine the minimum SOP and POS expressions for the function $f(x_1, x_2, x_3, x_4) = \sum m(4,6,8,10,11,12,15) + D(3,5,7,9)$

EXERCISES

Q. Draw the Karnaugh map and determine the minimum-cost SOP and POS expressions for the function $f(P,Q,R,S) = \sum m(0,2,5,7,8,10,13,15)$

Q. Draw the Karnaugh map and determine the minimum-cost SOP and POS expressions for the function $f(A,B,C,D) = \sum \Pi(3,5,7,8,10,11,12,13)$

EXERCISES

Q. Write an SOP expression for this truth table, and then draw a gate circuit diagram corresponding to that SOP expression

Then, simplify this expression using Boolean algebra techniques, and draw a simplified gate circuit diagram based on the final reduced expression.

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

EXERCISES

Q. Examine this truth table and then write both the SOP and POS standard Boolean expressions describing the output.

Which of those Boolean expressions is simpler for this particular truth table? Which will be easier to reduce to simplest form (for the purpose of creating a gate circuit to implement it)?

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

EXERCISES

Q. Write a POS expression for the output corresponding to this truth table

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

EXERCISES

Q. Write an SOP Boolean expression for this truth table, then simplify that expression as much as possible, and draw a logic gate circuit equivalent to that simplified expression.

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

EXERCISES

Q. Write an SOP Boolean expression for this truth table, then simplify that expression as much as possible, and draw a logic gate circuit equivalent to that simplified expression.

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

References

- E. Hughes, J. Hiley, K. Brown, and I. M. Smith, “Electrical and Electronic Technology,” Pearson Education Limited, England, 2008.

- A. Anand Kumar, “Fundamentals of Digital Circuits,” PHI Learning Private Limited, Delhi 2009.